DAO Governance Models for Social Impact and the Potential for a Human Rights Token for the Global Amnesty International Movement on the Cardano Blockchain

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Executive Summary

The core problem being addressed in this paper is that Amnesty International's current centralised and hierarchical organisational structure limits inclusivity, transparency, and efficient decision-making, hindering its ability to fully engage its vast global supporter base of 10 million members, particularly those facing freedom of expression restrictions and underrepresented voices from the Global South and Indigenous communities. To overcome these limitations, the proposed solution, as initially detailed in the Amnesty Human Rights DAO research paper, [1] is the creation of a Human Rights DAO (HRDAO). This paper outlines our mission: to create a decentralised supporter coordination ecosystem powered by smart contracts, tokenisation, and user governance on the Cardano blockchain, utilising a Human Rights Token to incentivise participation and create new value streams for advocacy, and the enhancement of human rights. Our ultimate aim is to foster an engaged, empowered, and transparent global human rights community capable of driving greater impact in education and action. Our proposed platform will enable frictionless community centric transactions, while giving users the power to shape the future of the network.

We can identify crucial opportunities for human rights and democracy empowerment through a blockchain HRDAO, and allow stakeholders, members and the general public to engage in the recognition of and respect for human rights, while contributing to resolving and highlighting current human rights issues.
A 2022 paper titled “Finding Web3’s soul” by leading strategists including Vitalik Buterin says:

“current Web3 is overly focused on transferable, financialized assets and lacks the primitives to represent social relationships and trust.” [2]

We respond to that critique by grounding the HRDAO in the lived realities of struggle and solidarity—shifting the focus from financialised assets to a framework of social trust, collective agency, and shared values become the foundation for digital coordination. For the first time, clusters in Amnesty International’s 10 million-strong global community can be brought into active alignment through blockchain—organised geographically or thematically across borders, and in deep solidarity with

those whose voices are suppressed behind closed ones. Through the Human Rights DAO, we aim to restore meaning to participation in the Amnesty network, and build a decentralised system where rights, dignity, and action are represented and actively reinforced by the architecture itself.

Relationality is key. This means engaging in conversations, fostering consensus equity, harnessing pluralism and collectively working towards measurable impact and progress. This includes in both building the DAO and token project, and enabling decentralised decision-making and shared responsibility in the DAO itself. As such, we treat optimal connections between people as a core objective of our approach.

What's at stake in the world of human rights is erasure, censorship, loss of memory and collapse of institutions. Rights dissolve in silence, and the architectures meant to safeguard dignity are eroded by bureaucratic inertia, technological force or digital amnesia. We also witness the disappearance of lived experience under the weight of opaque systems, and a slow unravelling of institutions that once held moral authority, while censorship mutates into subtler forms — invisibility, exclusion, algorithmic neglect. As Andreas Malm often warned, the infrastructures we rely on are not neutral; they carry histories of extraction and enclosure.
This project is about building something that holds memory and action together, so we're not just reacting to systems failure, but shaping something more resilient in its place. We are building human rights embedded futures.

We look forward to developing the exciting project further with both the Amnesty International, and Cardano and Project Catalyst communities.

This paper is being produced as a core output for Milestone 1 from the Project Catalyst funded Project 1300018 Amnesty International DAO and Human Rights Token Proof of Concept. [3]

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1 Introduction

"Web3 networks introduce a new type of internet-based institutional infrastructure that enables distributed internet tribes to self-organise and coordinate in a more autonomous way, steered by purpose-driven tokens, and executed with machine-enforceable protocols." [4]

This paper presents the proposed governance framework for the **Amnesty International Human Rights DAO (HRDAO)** and the **Human Rights Token (HRT)**. HRDAO aims to revolutionise human rights advocacy by integrating decentralised governance models within a secure, transparent, and equitable framework.

By leveraging Cardano's advanced governance infrastructure, HRDAO ensures:
- decentralised decision-making that empowers 10M+ Amnesty supporters;
- fair and accountable governance using reputation-based and quadratic voting models;
- tokenised incentive mechanisms to reward, upskill and take action;
- transparent funding distribution for grassroots human rights initiatives.

This paper conducts a **comparative analysis** of existing decentralised governance models and research to answer the following questions:

- *What are the key features of successful decentralised governance models in ensuring equity, transparency, and accountability?*
- *How do various models address power imbalances and ensure marginalized voices are included in decision-making?*
- *What metrics or indicators are most effective for measuring the impact of governance models on social justice or advocacy outcomes?*
- *What lessons can be drawn from failures in decentralized governance projects, particularly those involving large participation?*
- *Which governance mechanisms (e.g., quadratic voting, token-weighted voting, or reputation systems) are most appropriate for scaling human rights-focused initiatives?*
- *What alternate systems are being evolved to further define decentralized governance?*

We will be proposing the HRDAO's governance framework as multi-tiered and hybrid, integrating:
Delegated Representatives (DReps) for efficient decision-making
Quadratic Voting (QV) to prevent token-based plutocracy
Reputation-Based Voting to elevate grassroots activists' voices
AI-assisted governance tools to improve decision transparency
Participation Incentives and
Ethical Safeguards

With HRDAO 's Core Principles:

Decentralization → Power shifts from central bodies to the community
Transparency → All relative campaign funding and decisions are on-chain
Equity → Voting models prioritise grassroots participation

Inclusion → Reputation-based governance empowers marginalised and plural voices
Participation → Radical inclusiveness and rewards for engagement.

This paper provides a research-driven foundation for HRDAO's governance model and outlines its path towards tokenisation, and future impact leveraging Cardano's architecture. It also includes high-level goals, including social impact and the role of the novel HRT (Human Rights Token).

Methodology:

Our methodology entails a review of the relevant texts and papers sourced from leading journals and platforms over the past 36 months, (or earlier if noted as seminal). Also stakeholder meetings, Web3 networking research and conference attendances, as well as Amnesty Stakeholder and Web3 Cardano industry feedback. This is an evolving experiment, not a turnkey model. To deliver on the project's true promise, we are mindful of critical humility, and any internal hesitations and real structural challenges in integrating blockchain into a global NGO. Further, we aim to Introduce clear **pathways to participation** for non-technical users; emphasising community co-design, multilingual and local nuance, and low-tech access.
As such our expanding network of stakeholders, advisers and supportive parties is a crucial resource to keep the project grounded and malleable to the ever evolving tech and human rights landscape. We recognise the dual role of Amnesty as a legacy organisation with 65 years of human rights experience, and as an experimental actor in the nascent world of blockchain and real-world social impact.

Commentary:

The Internet is rapidly evolving, and with the advent of Web3, we now have the opportunity to reimagine how digital assets and services are structured. The convergence of decentralised technologies and global advocacy provides a groundbreaking opportunity to reimagine governance systems and funding mechanisms for human rights initiatives. Our platform aims to empower individuals by putting control back into their hands, allowing them to create, govern, and participate in an empowering decentralised network. At the forefront of this innovation lies the concept of a Human Rights DAO and Token, blockchain-based tools designed to amplify grassroots advocacy, ensure transparency, and empower global participation.

DAOs *"leverage the borderless nature of digital coordination, assembling a diverse group of people across the world, not only to facilitate change, but also to give birth to new, vibrant initiatives and entities that reflect their collective ideas and skills."* [5]

We recognise the **urgency and gravity** of the political moment: genocide, surveillance, censorship, collapse of civic trust and a general erosion of human rights worldwide make this project timely and impactful as we build new systems of coordination and cooperation towards human rights embodied futures. Coupled with these is the general geopolitical context of convergent structural crises: defunding of aid organisations, prolonged illegal warfare, the rise of digital authoritarianism and state-platform data mining collusion to name a few. Our Human Rights DAO and Token (HRDAO) project is a response to these contexts, not just a governance experiment.

Project Catalyst-funded and incubated within the Cardano ecosystem, HRDAO leverages the network's advanced governance capabilities – multi-asset native tokens, quadratic voting, on-chain treasury logic, and CIP-1694 compliant consensus models. Yet it does so with an eye to the future:

its architecture is designed to remain tech-agnostic, interoperable, and ultimately answerable not to the constraints of any chain, but to the movements it aims to serve.

Amnesty International ("Amnesty") is a non-governmental organisation focused on human rights protection, with a vision of a world in which every person enjoys all of the human rights enshrined in the Universal Declaration of Human Rights and other international human rights instruments. Since Amnesty International's founding in 1961, the organisation has uncovered human rights abuses, campaigned to mobilise the public and put pressure on governments to end the practices and change the systems that create them. Amnesty has grown to become the world's leading human rights organisation, with over 10 million members globally who support and enable the work of Sections advocating in over 60 countries. It have been able to campaign globally for the rights of prisoners of conscience around the world, the establishment of the International Criminal Court, the acknowledgment of women's rights as fundamental human rights, and an international framework to protect refugees, displaced people and undocumented migrants, among many other human rights issues that continue to evolve and expand.
The **Amnesty International DAO and Human Rights Token** (HRDAO + HRT) project aims to establish a decentralised governance framework that empowers this Amnesty International's global membership base of over 10 million people.

This paper outlines the governance framework for the proposed HRDAO and HRT, a dual-token system designed not for speculation but for solidarity. HRT-GOV is a non-transferable, soulbound governance token earned through real-world advocacy and verified contribution. HRT-UTL is a fungible utility token for campaign access, peer recognition, and micro-rewards. Together, these tokens operate not as currency, but as **relational proof** of care, action and presence.

We draw from multiple sources: the failures of past DAOs plagued by plutocracy and rug pulls; the theories of commons governance and social currency from Elinor Ostrom to David Graeber; the lived knowledge of Amnesty members on the ground in Gaza, Taipei, Bangkok, and Aotearoa New Zealand. The DAO's design integrates:

- Quadratic voting and delegated representatives (DReps) to decentralise voice,
- Human-curated oracles for verifying activist contributions and off-chain engagement,
- Privacy-first tools inspired by Telegram, Tor, and the Internet Archive,
- Memory-preserving infrastructures using decentralised storage, and zK-ID layers,
- Multilingual governance access to dismantle linguistic colonialism,
- A burn-to-signal and ritual token economy that anchors value in action, not speculation.

This project begins with political realism. We know token models alone will not end state repression or dismantle surveillance systems. But we also know that governance infrastructure matters, and that how we build it determines who can belong. If Web3 is to mean anything, it must mean a decentralised right to organise, to speak, and to care. If Amnesty is to survive and thrive in a decentralised participatory world, it must centre the same.
The HRDAO is not just a governance tool – it is **an infrastructure of memory and mutual protection**, designed for a collapsing civic reality.
Built with Cardano's funding and tooling, but structurally tech-agnostic, the DAO aims to enshrine human rights participation into digital systems that resist surveillance, censorship, and extractive logic.

In a time when Gaza's journalists are targeted, Sudan's internet is cut, and Amnesty's external civic channels are squeezed by funding cuts and platform censorship, we cannot build neutral systems. We must build **ethical, decentralised counter-institutions** that honour action, preserve testimony, and protect those who risk everything to speak.

The initial phase of this paper centres on identifying and analysing governance mechanisms such as quadratic voting, reputation systems, and token-weighted voting. These approaches are evaluated for their scalability, ethical integrity, and alignment with human rights objectives. Complementing this analysis is a detailed exploration of tokenomics challenges, including incentive issues, the risk of plutocracy, and mitigation strategies to safeguard mission-driven initiatives.

Key deliverables from this paper include:

1. **Comparative Analysis:** A review of existing DAO governance models, highlighting successes, failures, and their applicability to Amnesty's human rights advocacy. This also includes a focus on tokenomics challenges and mitigation strategies.
2. **Use Case Development:** Scenarios demonstrating the practical applications of the Human Rights Token that align with Amnesty's mission and are actionable within campaign strategic deliverables, including demonstrating how the DAO and token design would work in practice.
3. **Strategic Roadmap:** A phased implementation plan that aligns with Amnesty International's advocacy goals while addressing technical, ethical, and operational challenges. This is included in a seperate Project Plan document.

By synthesising insights from blockchain experts, human rights advocates, and Amnesty stakeholders, this milestone lays a foundation for actionable solutions that merge technological innovation with social impact.

2 Problem Statement

Amnesty International faces critical challenges in transparency, accountability, and supporter coordination across its 10-million-strong global network. Traditional hierarchical structures limit participation, particularly for those in the Global South and in contexts of digital authoritarianism, state censorship, or postcolonial aid dependency; and Amnesty lacks an effective mechanism to engage and coordinate supporters across different sections, making it difficult to leverage crowd-sourced knowledge and to respect the decentralised lived-experience intelligence from diverse activist ecosystems.
Amnesty International's mission is to ensure that every person enjoys the full spectrum of human rights enshrined in the Universal Declaration of Human Rights and other international human rights instruments.

Its work is grounded in principles of justice, dignity, and accountability, with key focus areas including:

- **Freedom of expression & political rights** – Defending activists, journalists, and political prisoners.
- **Abolition of torture & the death penalty** – Campaigning against inhumane treatment and executions.

- **Refugee and migrant rights** – Advocating for protection and fair treatment of displaced persons.
- **Gender, racial, and social justice** – Fighting discrimination and promoting equal rights.
- **Corporate and state accountability** – Holding governments and businesses responsible for human rights violations.

Amnesty operates through research, grassroots activism, legal advocacy, and international campaigning, leveraging its 10-million-strong supporter base to create systemic change. At its core, Amnesty envisions a world where human rights are respected, protected, and fulfilled for all, without discrimination.

Amnesty's existing governance architecture reflects a legacy of centralised control that unintentionally mirrors the same colonial structures it often campaigns against. The organisation would greatly benefit from a transformative solution that democratises decision-making, streamlines operations, and enhances impact measurement. The absence of a decentralised, transparent engagement platform means that Amnesty cannot fully utilise the potential of its global supporter base—an untapped resource with the potential diplomatic and strategic weight of a nation. The HRDAO turns supporter momentum into a form of programmable soft power.

We also acknowledge the differential ability to participate based on surveillance, bandwidth, local repression and the general geographies of data exclusion—aspects of the human rights struggle Amnesty campaigns on every day. However, we don't assume that more voices equals better outcomes. This has been shown (in DAO failures) to result in governance fatigue, vote capture, and fragmentation unless accountability mechanisms and trust systems are built in.
According to Ding et al. (2023), DAOs benefit from hybrid voting systems calibrated along axes of Security, Efficiency, Effectiveness, and Decentralisation (SEED).
HRDAO adopts this framework by layering:

- Quadratic voting to dilute plutocratic influence;
- Conviction voting to anchor genuine commitment;
- Reputation and knowledge-based weights to elevate practitioner voices; and
- Multi-language and multi-perspective capture for scalable deliberation.

This design balances on-chain impact with mission coherence, enabling rights defenders to govern, with knowledge, care, and collective witness.

"We don't need to be citizens of nation-states with defined territories and the rights and duties that traditional citizenship entails. Instead, we can move between 'digital tribes' and 'network archipelagos,' which may exist anywhere, at any time." [6]

These "digital tribes" are not neutral—they are shaped by access, power, and code. Our DAO must be rooted not only in protocol, but in people, place, and plural histories: a decentralised solution for decolonised futures.

3 The Solution

The proposed solution is a **Cardano blockchain-based Decentralised Autonomous Organisation (DAO)** designed to enhance Amnesty International's supporter coordination, transparency, and decision-making. By leveraging **Cardano's secure, scalable, and energy-efficient blockchain**, and integrating smart contracts and tokenisation, along with an integrated stack of technical complements, the HRDAO will incentivise positive human rights engagement, create a transparent governance framework, and allow supporters to record micro-actions securely on an immutable ledger. The HRDAO on Cardano is not just a platform, but a participatory protocol, and allows individuals, especially those long excluded, to shape not just outcomes, but the very terms of participation. In this DAO, rights are not passively defended – they are collectively encoded, acted upon, and verifiably protected.

This approach not only aligns with Amnesty's mission but also enhances digital democracy, ensuring that marginalised voices are heard.

Key Features:

- **Decentralised Governance** – Supporters participate in a democratic voting system using Cardano's on-chain governance mechanisms to make transparent funding and advocacy decisions.
- **Human Rights Tokenisation** – A value-backed token rewards engagement, incentivising actions like advocacy, education, and participation in campaigns.
- **Dual-token Model:**
  HRT-UTL (utility, transferable): for light interactions, education, campaign tools.
  HRT-GOV (soulbound): for governance, reputation, and long-term contribution.
- **Censorship-resistant Activism** – On-chain documentation of human rights abuses.
- **Transparent Impact Tracking** – Using Cardano's ledger, all supporter activities and funding distributions are publicly verifiable, ensuring accountability.
- **Smart Contracts for Efficiency** – Automated execution of decisions and resource allocation via **Plutus smart contracts** reduces administrative bottlenecks.
- **Conviction Staking** and quadratic-weighted thresholds to ensure the DAO voice doesn't scale with capital.
- **Inclusivity & Secure Identity** – Using Cardano's Atala PRISM identity solution, and Midnight's zK proofs, supporters, including those in restricted regions, can securely participate while maintaining privacy.
- **A Trust, Safety and Consent layer** to ensure participants can opt-in and including mechanisms for dynamic consent (ability to change over time), encrypted participation and proxy representation (for high-risk zones).

By integrating these elements, the HRDAO transforms Amnesty into a borderless, participatory movement with diplomatic-level influence, ensuring human rights advocacy is collective, transparent, and action-driven, creating new forms of value exchange aligned with Amnesty's mission. The HRDAO is not a platform built *for* supporters, but a governance process built *with* and *by* them, with local, thematic, and linguistic nodes empowered to evolve their own tools and tactics.

By decoupling participation from capital and prioritising time-, knowledge-, and care-based contributions, we ensure a token design that honours human dignity over speculative gain.

The project acknowledges the need for careful planning and validation to ensure the feasibility and alignment of a decentralised governance model with Amnesty's mission. The initial research and conceptualisation phase directly tackles this by reviewing common problems and by exploring successful decentralised governance models.

Ensuring the long-term sustainability and impact of these decentralised mechanisms is another key concern. The tokenomics modelling and simulation phase is specifically designed to address this by reviewing various models for token distribution, rewards, and governance influence. Key here is a natural balance between the token supply and demand factors.

The project also focuses on the practical challenges of building and deploying a decentralised platform. The MVP (Minimum Viable Product) development and testing phases aim to create a functional prototype and gather feedback from stakeholders to refine its functionalities and ensure security and scalability. We will co-design the pilot with Amnesty activist communities in four countries – Taiwan, Thailand, Australia and New Zealand, while incorporating Amnesty stakeholder and human rights expertise and young digital native and Web3 supporters. The MVP aims to allow the pilot group to earn a token through verified advocacy and activism micro-actions, and burn it through proposal voting or other governance action-taking.

Finally, the project recognises the importance of measuring the impact and ensuring transparency and accountability throughout its lifecycle. The final milestone emphasises a detailed evaluation report and public documentation to share the project's outcomes with the Cardano community and Amnesty's network. The need for addressing legal and regulatory compliance in various jurisdictions is also identified as a crucial step for full public deployment, although this aspect along with security, scalability, localisation and community engagement/education are outside the scope of this paper.

4 Technical Overview

The HRDAO is designed to leverage the advanced governance and decentralisation capabilities of the Cardano blockchain, ensuring a transparent, efficient, and participatory framework for human rights advocacy. Cardano's Proof of Stake (PoS) consensus mechanism provides an energy-efficient alternative to traditional blockchain models, reducing environmental impact while maintaining high security and scalability. The Ouroboros consensus protocol further enhances decentralisation by ensuring that governance and transaction validation remain both secure and democratic. Additionally, Cardano's native token functionality allows seamless integration of the Human Rights Tokens (HRT), creating a mechanism for incentivising and rewarding activism and participation.

Quadratic and reputation-based voting systems reinforce democratic governance by prioritising activism and contributions over financial power. Within this framework, HRDAO members can propose and vote on Amnesty initiatives, allocate human rights funding, and earn governance tokens for verified activism.

By aligning with Cardano's SIP-1694 framework and the Voltaire governance model, HRDAO ensures long-term sustainability and deep integration within Cardano's governance roadmap, with

Amnesty's global members acting as the equivalent of ADA holders, and a Council of Stewards mirroring Cardano's Constitutional Committee.

Cardano's Voltaire governance system is a model for decentralised decision-making at scale. It introduces SIP-1694, which enables:

- Community-driven funding through the Treasury
- Decentralised Representatives (DReps) for governance participation
- On-chain voting for policy decisions.

How HRDAO Aligns with Cardano's SIP-1694 Framework:

HRDAO Governance Layer	SIP-1694 Counterpart
Amnesty Global Members	ADA Holders
Delegated Representatives (DReps)	SIP-1694 DReps
Council of Stewards	Constitutional Committee
Treasury Committee	Cardano Treasury Governance

Through **SIP-1694**, HRDAO can:
propose and fund human rights initiatives directly via Cardano governance,
leverage DReps to manage Amnesty's decentralised governance structure, and
secure long-term ADA funding via Cardano's treasury mechanisms.

The Cardano ecosystem is well-positioned to explore and enhance innovative governance solutions. Its design is grounded in a research-driven approach, aiming to balance decentralisation, scalability and sustainability. Through this integration, HRDAO secures the ability to propose and fund human rights initiatives directly through Cardano governance mechanisms, ensuring Amnesty's mission is not only decentralised but also sustainable.

Technical Architecture:

The **Amnesty International Human Rights DAO (HRDAO)** is an end-to-end, privacy-preserving, and governance-centric platform built on the Cardano blockchain, enhanced by Midnight's zero-knowledge layer and AI-driven agent support. This technical blueprint section details the HRDAO's modular architecture, user flow, governance model, tokenomics, privacy safeguards, and cross-chain interoperability. It serves both as a developer reference and a high-level guide for stakeholders and the board of Amnesty International. It also equips developers, activists, and stakeholders with a clear "recipe" to implement and scale the HRDAO—turning vision into verifiable impact.

HRDAO is composed of six interoperable layers:

1 User Interaction & Authentication

- **NuFi Single Sign-On (SSO):** Streamlined email/Google login, mobile-first wallet initialisation.
- **Atala PRISM DID:** Decentralised identity creation for compliance and KYC optionality.

- **NuFi Wallet Interface:** Responsive UX for wallet management, transaction signing, and token viewing.

2 **Identity & Privacy Layer**

4. **Mint zk-ID (Private):** Issue zero-knowledge identities via Midnight to preserve anonymity.

5. **Soulbound Tokens (SBTs):** Non-transferable credentials representing onboarding status and reputation.

6. **Advanced Encryption Standard (AES):** Encrypt metadata and off-chain evidence before storage.

3 **AI Companion & Campaign Module**

- **Activate MANUS AI Companion:** Personalised petition matching, campaign rotation, and action recommendations.

- **Featured Campaign:** Rotating global action with QR Code Check-In (DePIN) integration.

4 **Governance & Voting Layer**

- **Clarity Protocol (CIP-1694):** On-chain proposal lifecycle: submission, discussion, and voting.
- **Quadratic & Reputation Voting:** Prevents plutocracy; combines token-weighted and reputation metrics.

- **Delegated Representatives (DReps):** Elected delegates with enhanced voting power for protocol-level decisions.

- **Council of Stewards & Emergency Veto:** Ethical oversight council empowered to pause proposals.

5 **Smart Contract & Execution Layer**

- **Aiken Smart Contracts & Plutus Scripts:** Business logic for badge minting, fund disbursements, and triggers.

- **Midnight Private Contracts:** Confidential execution of sensitive proposals via zk-programs.

- **Cross-Chain Bridges:** EVM (Ethereum) and IBC (Cosmos) compatibility for multi-chain asset flows.

6 **Storage, Oracles & Feedback Loop**

- **IPFS / Arweave / Filecoin:** Decentralized storage for evidence, art, and testimony with content addressing.

- **Oracles & Impact Metrics:** On-chain data feeds for real-world impact assessments and SDG alignment.

- **Feedback Loop:** Automated reports, dashboards, and mobile notifications to close the user flow.

User Flow Breakdown:

Onboarding: NuFi SSO → Create Wallet (Mobile-First UX) → Mint zk-ID → Issue SBT-Badge.

AI Integration: Activate MANUS AI → Personalized Campaign Feed → Sign or Auto-Sign Petitions.

Engagement: Join Protests via QR Code Check-In (DePIN) →Upload Evidence / Art / Testimony →Store on IPFS/Arweave/Filecoin.

Proposal & Voting: Access Clarity UI Dashboard → Submit / Review Proposals → Vote (Quadratic & Reputation) → See Real Time Voting Outcome.

Execution: Trigger Smart Contract → Automated Disbursement or Action → Mint Human Rights Badge.

Feedback & Rewards: Impact Dashboard → Earn Utility Tokens (HRT-UTL) → Build Reputation → Qualify for Governance Tokens (HRT-GOV).

Dual-Token Model & Tokenomics:

- **Utility Token (HRT-UTL)**: Fungible, transferable; used for staking, campaign voting, micro-rewards, and access to services.

- **Governance Token (HRT-GOV)**: Soulbound (non-transferable); earned through verified contributions, required for protocol-level proposals, and council membership.

Key Economic Features:

- **Adaptive Supply:** Inflationary minting tied to participation metrics; governed by DAO consensus.

- **Staking & Vesting:** Time-locked yields for long-term engagement; vesting schedules for early contributors.

- **Velocity Controls:** Purpose-driven sinks via proposal fees and service access to manage circulation.

- **Treasury Mechanisms:** Catalyst grants, quadratic crowdfunding, multi-sig vaults, and emergency veto triggers.

Security, Privacy & Ethical Safeguards:

- **Zero-Knowledge Proofs:** Anonymous credentialing and private voting with zk-circuits.

- **AES Encryption:** Secure off-chain data and evidence storage.

- **Ethical Constitution:** Immutable Amnesty DAO Charter and Human Rights Impact Assessments for proposals.

- **Whistleblower & Reporting Protocols:** Protected channels with zk-anonymity and multi-stakeholder review.

Cross-Chain & Scalability:

- **EVM & Cosmos Bridges:** Asset portability and interoperable DePIN integration.

- **DAO-as-a-Service Templates:** Modular deployment for regional or partner DAOs (e.g., Climate Justice DAO).

- **Localization & Accessibility:** i18n support, PWA design, and voice-accessible interfaces for global inclusion.

HRDAO represents a covenant between global human rights defenders and the technology that serves them. By embedding decentralised governance, regenerative economics, and ethical safeguards into Cardano's and Midnight's infrastructure, HRDAO transcends traditional organisational models, engineered for trust, dignity, and collective action.

We also recognise the need for a crucial **Embodied Governance layer** for the HRDAO. This forms a cultural-ethical layer for consent, legitimacy, and inclusion. Beyond smart contracts, the HRDAO operates on *social contracts*. These embodied governance rituals include consent protocols, language justice, cultural mediation, and care-driven onboarding. They ensure the DAO flourishes with human dignity and pluriversal sensibilities at its core.

This layer would support:
- Cultural interpretation and consent frameworks
- Decentralised onboarding rituals (e.g., digital circles, storytelling-based reputation)
- Acknowledgement and incorporation of Indigenous and Cultural data sovereignty frameworks. [7]
- Knowledge of feminist and postcolonial tech critiques.
- Centering localised sense-making, and relational ethics.

Technical systems often abstract governance into votes, tokens, or contracts. Yet real-world legitimacy – especially in human rights and activist contexts – emerges from rituals, relationships, and shared cultural meaning. Current DAO structures tend to ignore embodiment, storytelling, care work, and informal authority structures.

Core components of this layer would include:

1. Cultural Interpretation & Consent Frameworks

Our governance must recognise the diversity of how consent, authority, and participation are understood across different cultural, legal, and spiritual contexts.

- Localised Ethical Protocols: Each regional HRDAO node can define its own consent practices (e.g., collective consent, elder validation, silent affirmations).

- Dynamic Consent Tools: Allow participants to update or revoke data use, governance role, or token visibility at any time.

- Cultural Consent Prompts: When engaging, users could opt into worldview-aligned governance frames (e.g., Ubuntu, adat, Islamic legal ethics).

Example: An Indigenous community may opt into a consensus model rooted in *circle practice* instead of majority vote.

2. Decentralised Onboarding Rituals

We will move away from extractive user acquisition toward participatory and embodied community welcome processes.

- Digital Story Circles: Before accessing governance, contributors attend (or listen to) decentralised storytelling rounds where members share values, origins, and resistance histories.

- Narrative-Based Reputation: Instead of raw activity points, users accumulate "narrative badges" based on stories witnessed, told, or acknowledged.

- Ethical Orientation Modules: Instead of "tutorials," participants undergo orientation experiences (e.g. decolonial workshops, human rights advocacy skills, digital safety training).

Example: Upon joining the HRDAO, a user might be invited to share their advocacy story in a secure audio log, or a specific unique cultural forum that earns them a soulbound onboarding badge, or redeemable utility tokens.

3. Guardianship Protocols for Risk Zones

Not all users can or should participate equally due to risk - whether they are refugees, political prisoners, or activists in authoritarian contexts.

- Guardian Delegation: Trusted proxies (e.g., diaspora advocates, NGOs) can co-sign or shield votes on behalf of at-risk participants.

- Zero-Knowledge Co-Governance: Using zK-proofs to allow anonymous but verifiable participation from repressive zones.

- Conditional Visibility: Allow participants to toggle their public presence in governance – visible in solidarity zones, masked in hostile ones.

- A dignity imbued protocol that allows for the right to vanish, layered consent (allowing users to participate at multiple levels without exposing all metadata), and Proxy or guardian action.

Example: A human rights lawyer in Iran can vote anonymously, with a diaspora guardian able to confirm their input without revealing their identity.

This layer grounds HRDAO in dignity and shared care, not just cryptographic finality. It moves governance beyond code into respect for culture and cultural safety. We believe governance must be embodied, relational, and context-aware. HRDAO refuses the logic of the singular system. Instead, we cultivate a pluriverse of governance rituals — shaped by ancestral memory, present urgency, and the imaginative resilience of those on the margins.

5 Governance Model & Comparative Analysis

Research findings:

We conducted a comprehensive analysis of available research resources from the period 2022-2025, using SSRN, Science Direct, arXIV, Frontiers Media, Github, Stanford Research, as well as Cardano eco-system resources like Input | Output, Emurgo, Cardano Spot and Cardano Foundation, and here present key insights about DAO governance challenges and corresponding strategic recommendations, and a synthesis of these ideas into a proposed Governance Model.

Here we can directly answer one of the research questions:
What lessons can be drawn from failures in decentralised governance projects, particularly those involving large participation?

Challenges of DAO models:

1. Centralisation of Power

Despite the promise of decentralisation, many DAOs struggle with concentrated decision-making power. Research indicates that governance is often dominated by a small number of large token holders ("whales"), with some DAOs having as few as three or four stakeholders determining vote outcomes (Falk et al., 2024). This centralisation undermines the fundamental purpose of DAOs and raises questions about their legitimacy as truly decentralised organisations. Sharma et al. (2024) found that DAOs with tradable governance tokens consistently exhibit more centralised decision-making structures, while Kitzler et al. (2023) revealed that many DAOs claiming decentralisation are, in fact, controlled by a small group of contributors or insiders. This concentration of power not only contradicts the philosophical underpinnings of DAOs, but also creates practical vulnerabilities where a few actors can potentially manipulate governance outcomes to serve their interests rather than the broader community's needs. Decentralisation is a key bedrock to the success of the coordination of the Amnesty worldwide movement.

2. Ethics-by-Design

Decentralised systems need not just fairness metrics but foundational values that resist co-option, protect the vulnerable, and encode solidarity. Without embedded ethical principles, DAOs risk being hijacked by malicious actors or having their priorities shifted away from community values. Research indicates that hardcoded ethical safeguards are necessary to ensure alignment with human rights and community objectives (Kitzler et al., 2023). Han et al. (2024) emphasized the importance of embedding ethical safeguards within governance structures to align decision-making with community values and human rights principles. The ethical dimension of DAO governance extends beyond preventing explicit harm to actively promoting positive values and preventing the gradual erosion of community standards. DAOs operate in a global context with diverse stakeholders and must navigate complex ethical considerations across cultural boundaries. Without explicit ethical frameworks, governance decisions risk prioritising short-term financial gains over long-term sustainability or community well-being, potentially leading to outcomes that technically follow governance rules while violating the spirit and purpose of the organisation. Here we incorporate principles of Ostrom's community design (referred to in note 12 below), including where appropriate

regional or thematic nested decision-making, graduated sanctions and feedback loops to increase participant governance rights over time.
Crucial also to this challenge are considerations of the Right to Not Participate as a form of ethical consent. This reflects a commitment to consent, autonomy, and justice-centered design. Renowned social theorist Paulo Freire, in *Pedagogy of the Oppressed*, insists that education and participation must be dialogical, entered into voluntarily, and with critical consciousness. **The ability to abstain, to disengage, or to remain invisible** can also be an act of sovereignty.

DAO Application: Design Implications

Feature	Design Ethic
zk-ID & SBTs	**Must include revocation and expiration features for self-withdrawal.**
Onboarding	**Should offer an opt-out narrative explaining what not joining means, with no penalty.**
Governance Participation	**Reputation should not be damaged by absence or non-participation. Passive contribution (e.g., cultural witnessing, silent observation) should be valid.**
Metrics	**Impact tracking must differentiate between absence due to disengagement and *conscious resistance* (e.g., boycotts, silent protest).**

Participation in HRDAO is not a mandate — it is an invitation. We honour the right to engage, the right to retreat, and the right to remain unseen. We recognise the need to be able to protect whistleblowers, frontline defenders, undocumented persons, and those under surveillance with dignity-first design. The HRDAO affirms the Right to Not Participate as a foundational principle of ethical consent.

3. Low Voter Participation

Voter engagement in DAOs is alarmingly low, with only 3-7% of token holders typically participating in governance votes (Falk et al., 2024). This low turnout not only questions the legitimacy of decisions but also correlates with increased centralisation as power defaults to the most active stakeholders, who are often the largest token holders. Sharma et al. (2024) established a direct correlation between low voter participation and increased centralisation, creating a negative feedback loop where centralisation further discourages participation from smaller stakeholders who feel their votes won't matter. Ballandies et al. (2024) identified declining participation as one of the key challenges facing DAOs over time. This participation crisis undermines the core value proposition of DAOs as community-governed entities and threatens their long-term viability as governance mechanisms. Without broad stakeholder involvement, DAOs risk becoming oligarchies masquerading as democratic institutions. Successful models place grassroots participation at their core, designing incentive systems, narrative-based onboarding, and reputation-weighted voting to ensure that legitimacy arises from broad, inclusive engagement, not from concentration of power among a few token- or reputation-rich actors.

4. Token Value and Decentralisation Paradox

Higher token market values often correlate with lower levels of decentralisation (Sharma et al., 2024). This creates a paradox where financial success can actually undermine governance objectives. Research shows that social or public-good focused DAOs tend to maintain better decentralisation, while infrastructure and investment DAOs show greater centralisation over time (Sharma et al., 2024). This pattern suggests that as tokens become more valuable, large stakeholders have greater financial incentives to maintain or increase their influence, while smaller holders might be tempted to sell their governance rights to profit from appreciating tokens. The result is a governance structure that becomes increasingly plutocratic as the protocol gains market value—precisely the opposite of what many DAOs intend to achieve. This paradox forces DAOs to grapple with fundamental questions about aligning financial incentives with democratic governance principles. Successful models incorporate tiered, resilient architecture (for instance DReps, Council of Stewards, CIP-1694 alignment), and map decision types to layers of authority, reputation or regional expertise.

5. Voting Mechanism Limitations

Traditional token-based voting systems tend to reinforce power imbalances, as they give disproportionate influence to those with larger holdings. Alternative mechanisms like quadratic voting and conviction voting each have their own strengths but also face implementation challenges such as vulnerability to Sybil attacks (Shah, 2024). Token-based voting essentially creates a plutocracy where wealth translates directly into governance power, while quadratic voting attempts to balance influence but requires robust identity verification to prevent gaming of the system. Although quadratic voting is vulnerable to Sybil attack if identity is not strong (with introduced zk-DID for instance). Conviction voting reduces last-minute manipulation by accumulating voting power over time but may still favour wealthier participants who can afford to lock up their tokens for extended periods (Shah, 2024). Reputation systems must also avoid social credit pitfalls — gamification or penalisation of dissent.

The search for an ideal voting mechanism highlights the tension between efficiency, security, and equitable representation that all DAOs must navigate, with no single solution emerging as universally optimal for all contexts.

6. Transparency and Documentation Deficiencies

Many DAO proposals lack clear descriptions and documentation, hindering informed decision-making (Ma et al., 2024). This opacity makes it difficult for community members to understand the potential impacts of their votes and reduces overall participation quality. Ma et al. (2024) found that a significant percentage of proposals across multiple DAOs lacked adequate descriptions, creating information asymmetries between insiders who understand the context and average community members who don't. Laturnus (2023) observed that while transparency may enhance governance processes, it doesn't necessarily translate to better financial performance, suggesting a complex relationship between governance visibility and outcomes. Poor documentation practices create barriers to entry for new participants and enable information arbitrage opportunities for insiders, further concentrating power among those "in the know." Clear and accessible information is the foundation of informed voting, and without it, governance decisions risk becoming exercises in blind trust rather than collective wisdom.

7. Governance Token Manipulation

Research has identified concerning patterns of governance token ownership changing shortly before important votes, suggesting potential vote-buying or manipulation (Kitzler et al., 2023). Additionally, some DAOs distribute governance tokens in ways that mirror traditional equity offerings, raising regulatory concerns (Falk et al., 2024). These manipulation risks threaten the integrity of governance processes and undermine trust in DAOs as fair decision-making systems. Kitzler et al. (2023) documented suspicious patterns of token movement before critical votes, while Manzano Kharman & Smyth (2024) highlighted how small to mid-sized token holders are economically incentivised to abstain from voting or sell their votes, creating markets for governance influence. This manipulation may be particularly problematic in DAOs structured as regulatory shields, where centralisation through token manipulation has led to regulatory intervention, as demonstrated in the Ooki DAO case (Falk et al., 2024). These vulnerabilities highlight the need for governance systems that can resist economic incentives for vote manipulation.

8. The Free-Rider Problem

DAOs face a classic economic challenge where participants can benefit from organisational success without actively contributing to governance or operations, leading to governance inefficiencies and unfair distribution of responsibilities (Laturnus, 2023). This problem is exacerbated in blockchain governance where governance work incurs real costs in the form of transaction fees and opportunity costs, yet the benefits of good governance are distributed to all token holders regardless of their participation. The economic disincentives for fair voting noted by Manzano Kharman & Smyth (2024) further compound this issue, as transaction fees discourage voting, particularly for small to mid-sized token holders. Over time, this free-rider problem can lead to governance fatigue among active participants who bear disproportionate responsibilities, potentially resulting in governance collapse if not properly addressed. Creating sustainable governance requires carefully designed incentive structures that align individual benefits with community participation.

Lessons from Failed DAOs & HRDAO's Approach to Governance:

Decentralised governance projects with large participation have faced critical failures, often due to security flaws, low engagement, and governance imbalance. HRDAO applies these lessons to create a resilient, equitable, and scalable governance model that prioritises activist participation over financial influence.

One of the most infamous failures, The DAO (2016), suffered from smart contract vulnerabilities, leading to a $50M hack and Ethereum's hard fork. HRDAO mitigates such risks by prioritising audited smart contracts and security-first development. Similarly, FEI Protocol DAO collapsed due to economic design flaws, showing that even robust governance cannot compensate for unstable financial models—a lesson HRDAO addresses by designing a well-structured Human Rights Token economy with compliance safeguards.

Additionally, AragonDAO (2023) dissolved due to internal conflicts between token holders and core contributors, highlighting the need for governance that balances decentralisation with operational control. HRDAO avoids governance deadlocks by integrating Delegated Representatives (DReps), ensuring expert-led decision-making while maintaining decentralisation.

To foster long-term participation, HRDAO adopts mechanisms from successful DAOs like GitcoinDAO's quadratic funding (ensuring equitable decision-making), NounsDAO's continuous funding model (ensuring sustainability), and dxDAO's resilience through decentralised governance. By combining quadratic voting, reputation-based governance, and DReps, HRDAO builds a governance model that is transparent, participatory, and resistant to manipulation.

Building on the **Decentralisation Paradox** mentioned in point 3 above, **Vili's Paradox**, developed by economist **Vili Lehdonvirta**, highlights a fundamental governance dilemma in distributed ledgers. If such systems rely solely on voluntary agreement to resolve disputes, they risk failure due to deadlock or inaction. However, when they introduce informal governance structures to manage decision-making, they compromise true decentralisation.

As Werbach (2018) observes,
"the real success of blockchain-based systems will depend on their internal capacity to instantiate new forms of governance." [8]

While blockchain-based organisations like DAOs can reinforce governance mechanisms embedded in their technical architecture, this does not necessarily extend to broader decision-making processes—such as rule enforcement, initiative advocacy, policy creation, or the strategic architecture of campaigns and fundraising efforts. We reach the same conclusion as Werbach where he goes on to state that blockchain systems:

"will be subject to overlapping governance structures, with varying degrees of centralization." [9]

The outline further below reflects this approach.

Further, a comprehensive survey of DAOs conducted in 2021, [10] which included 422 survey respondents from 223 DAOs in 290 cities around the World, found DAO governance faces challenges such as lack of clear guidance, shallow decentralisation, and reliance on coin-based voting, which can centralise power and limit inclusivity. To improve, DAOs should implement better coordination tools that reward consistent contributors, move beyond pure token-weighted voting to more equitable decision-making models, and foster deeper community engagement. Ensuring transparent communication, and fairly distributing governance power will strengthen DAOs, making them more resilient, participatory, and aligned with their core missions, including plural, collective intelligence. The outline below also reflects relevant aspects of these findings.

Key Features of Successful Governance Models:

Successful decentralised governance models share three essential qualities:

Feature	Why It Matters	Example DAO
Accountability Mechanisms	Prevents fraud &	Gitcoin Grants (Public Goods

Feature	Why It Matters	Example DAO
	corruption	DAO)
Transparent Voting & Decision-Making	Increases trust & participation	ENS DAO (Ethereum Name Service)
Equitable Participation	Avoids power imbalances	Cardano's SIP-1694 Model

These principles inform HRDAO's reputation-based & quadratic voting model.

Metrics for Measuring Social Justice Impact:

HRDAO uses **on-chain analytics** to measure its impact:

Metric	Measurement Approach
Proposal Inclusivity	Number of marginalised group proposals funded
Funding Transparency	Smart contract-based treasury allocation tracking
Voting Participation	Unique voter count per governance cycle
Advocacy Impact	Measurable policy changes achieved

These metrics ensure HRDAO remains mission-aligned & data-driven.

Decentralisation Spectrum – Tiered Nature

The evidence suggests that DAOs are more likely to reflect a spectrum of decentralisation rather than a binary fully decentralised or centralised structure. Striking the right balance is essential for DAOs to achieve their goals effectively. Our proposed tiered structure is a response to this, by providing a **Council of Stewards** DAO role that ensures key structural decisions are able to be made by Amnesty stakeholders and partners ensuring continual alignment with the Amnesty mission.

This echoes the concept of an 'emergency regime' – which
"assigns a narrow set of powers to a predetermined group of delegates to frame approaches to resolving an exception, which network actors will then consider and implement." [11]

Successful DAO Governance Examples

- **GitcoinDAO** – Quadratic funding model & public goods funding
  1. Introduced quadratic voting for governance to balance power.
  2. Success in distributing millions to public goods projects with minimal governance friction.
- **NounsDAO** – Continuous funding model & meme-driven governance
  1. Uses auction-based funding for governance sustainability.
  2. Encourages experimentation while avoiding typical plutocracy traps.

- **dxDAO** – Governance participation & resilience
    1. Focuses on decentralised decision-making with token-weighted voting.
    2. Successfully funds and maintains DeFi projects.

Failed DAO Governance Examples:

- **The DAO (2016)** – Smart contract failure & governance failure
    - Lack of effective governance oversight led to a $50M hack and Ethereum's infamous hard fork.
    - **Lesson:** Code-based governance without rapid adaptability can be a fatal flaw.
- **FEI Protocol DAO (2021-2022)** – Failed governance proposals & economic collapse
    - Attempts to stabilise an algorithmic stablecoin through governance voting failed due to lack of consensus.
    - **Lesson:** Economic design flaws can overshadow governance structures.
- **AragonDAO (2023)** – Governance conflicts led to DAO dissolution
    - Internal governance struggles between token holders and core contributors led to Aragon's reset.
    - **Lesson:** Governance needs to balance decentralisation with operational control.

Case Studies: Quadratic Voting & Reputation-Based DAOs

1. Case Study: Gitcoin Grants & Quadratic Funding

Gitcoin Grants is a well-known experiment in **Quadratic Funding (QF)**, a mechanism where contributions from individual donors are matched from a shared funding pool. The match is determined using quadratic voting principles, ensuring that broad support matters more than large donations.

Why Quadratic Funding?

Traditional funding models suffer from plutocracy, where large donors dictate outcomes. Quadratic funding amplifies the voices of smaller contributors, making grassroots initiatives more viable.

How It Works

- Users donate to projects they support.
- A matching pool is allocated using a quadratic formula: funding donations - Total funding=∑individual donations
- This formula ensures a project with 100 small donations gets more funding than one with a few large donations.

Impact & Metrics

Metric	Gitcoin Results
Total funds distributed	$50M+
Participating projects	3,000+
Unique donors	100,000+

Lessons for HRDAO

Quadratic Voting (QV) could ensure diverse participation in HRDAO decision-making.
Quadratic Funding (QF) could help distribute treasury funds more equitably to grassroots human rights initiatives.
Ensuring Sybil resistance (i.e., preventing fake accounts) is critical—Gitcoin uses Gitcoin Passport (decentralised identity – now Human.Tech) to verify real contributors.

2. Case Study: Proof-of-Humanity & Reputation-Based DAOs

Proof-of-Humanity (PoH) is an identity-verification system that ensures only real, verified humans participate in governance. It is used in Kleros Court, a decentralised dispute resolution DAO, and for Universal Basic Income (UBI) token distribution.

Why Reputation-Based Governance?

Token-weighted voting in DAOs often leads to wealth concentration and whale dominance.
Reputation-based models reward users based on their contributions, not their financial holdings.

How It Works

- Users register with Proof-of-Humanity (verified by other humans).
- Reputation scores are built over time through participation.
- Governance weight is determined by contributions, not token ownership.

Impact & Metrics

Metric	Proof-of-Humanity Results
Verified human users	20,000+
UBI distributed	5M+ UBI tokens
Dispute resolutions in Kleros Court	1,500+

Lessons for HRDAO

HRDAO should integrate reputation-based governance to empower activists rather than wealthier participants.
On-chain identity verification (DIDs) paired with zK proofs could prevent governance manipulation.
Governance power should increase based on long-term participation, rather than financial holdings.

Key Takeaways for HRDAO

Feature	Gitcoin (Quadratic Funding)	Proof-of-Humanity (Reputation-Based Governance)	AI-DAO Implementation
Voting Power	Quadratic formula	Reputation-based	Hybrid: Quadratic & Reputation-Based Voting
Sybil Resistance	Gitcoin Passport	PoH Verification	Decentralised ID (DID) and zK Integration
Equity	Amplifies small voices	Rewards long-term participation	Ensures activists & grassroots communities have a voice
Governance Risks	Fake identities, collusion	Verification overhead	HRDAO uses DReps & smart contracts for moderation

The research presented in this section reveals both the promise and fragility of existing DAO governance structures. Patterns of low participation, centralised influence, and limited accountability raise crucial concerns for mission-driven ecosystems like Amnesty's. These findings shape the foundation for our next phase: a governance model rooted in inclusivity, cultural relevance, and real-world legitimacy. Informed by activist contexts and grounded in care, we now move toward a framework that values diverse contributions — including relational labour, peer validation, and narrative-based engagement — and integrates these into the architecture of decision-making.

6 Proposed Governance Model

Based on the above research, we can propose our own governance model for the HRDAO, and address another one of the research questions:
What are the key features of successful decentralised governance models in ensuring equity, transparency, and accountability?

Building on established DAO governance frameworks and the research above, it is critical to address one of the key challenges in decentralised governance—the ownership problem. Van Vulpen & Jansen (2023) highlight that many DAOs still rely on centralised infrastructure, such as cloud services or API providers, which creates vulnerabilities and risks of governance capture. To ensure true decentralisation, DAOs must integrate governance models aligned with Elinor Ostrom's principles for managing common-pool resources, [12] particularly those emphasizing stakeholder participation in rule-making, transparent decision-making, and effective monitoring mechanisms.

To ensure that the Amnesty DAO remains resilient against centralisation risks, its governance model must distribute authority, decentralise infrastructure dependencies, and embed accountability mechanisms. Van Vulpen & Jansen (2023) highlight that DAOs operating within centralised digital environments risk governance capture and infrastructural bottlenecks, ultimately weakening community autonomy. To counter this, Amnesty DAO must implement on-chain voting, multi-signature treasury controls, and decentralised identity verification to reinforce participatory governance. Additionally, adopting gradual decentralisation pathways, role-based governance, and transparent proposal mechanisms will ensure that decision-making remains inclusive, transparent, and resistant to coercion. These strategic recommendations not only align with Ostrom's principles of common-pool resource management but also provide a practical roadmap for ensuring long-term sustainability and democratic governance.

Strategic Recommendations:

1 A Multi-Tiered Hybrid DAO Model

Strategy: Create separate decision-making bodies with distinct responsibilities.

HRDAO is structured as a multi-tiered hybrid governance model, integrating decentralised decision-making with responsible stewardship. Unlike traditional DAOs that are either fully decentralised (e.g., Uniswap) or semi-centralised (e.g., MakerDAO), HRDAO balances community governance with structured oversight.
By separating governance functions, We can address the centralisation concerns identified by Sharma et al. (2024) while still maintaining efficient decision-making processes, potentially using alternative qualification methods like contribution history or reputation rather than token holdings.

For maximum effectiveness, we would want to eventually establish clear jurisdictional boundaries between roles and design escalation procedures for resolving disagreements. This structure acknowledges the reality that different stakeholders have different areas of expertise and concern, creating a more nuanced governance system than simple token voting can provide, and includes a combination of earned fungible tokens and non-fungible **Soulbound tokens** (non-transferable governance rights), [13] so that participants can gain more governance rights as more responsibility through proven advocacy and up-skilling is recorded (a reputation score).

Proposed Governance Layers of HRDAO

Layer	Role	Who Participates?	Powers & Responsibilities
Global Members (All Participants)	Community-level decision-making	Amnesty supporters, human rights activists, donors	Campaign voting, proposal feedback
Delegated Representatives (DReps)	Representing members in governance	Elected or reputation-based delegates	Governance changes, treasury priorities, dispute resolution
Council of Stewards/Treasury Council	Ensuring mission alignment Managing on-chain funding allocation	Experienced Amnesty leaders, blockchain governance experts Selected governance participants	Ethical oversight, fund allocation, mission alignment

This layered structure ensures HRDAO avoids the pitfalls of both fully centralised control (traditional NGOs) and unstructured DAO governance (which can lead to inefficiency and lack of direction). A combination of off-chain and on-chain governance in necessary.
We can also here note the possibility of introducing regional pods or "Civic Assemblies" beneath the

global member layer. Nested assemblies can deliberate locally or thematically while synchronising with the global HRDAO core, e.g., a Youth Justice Pod in Kenya, an Indigenous Rights Pod in Aotearoa New Zealand or a Stateless Citizens Assembly in Europe. These can autonomously submit proposals upward.

Voting & Decision Mechanisms:

- **Quadratic & Reputation Voting**: Balances power, prevents plutocracy.
- **Delegated Voting**: Liquid democracy enabling flexible delegation to DReps.
- **Emergency Veto**: Multi-stakeholder ethics panel can pause proposals.
- **Adaptive Quorums**: Variable thresholds based on proposal impact.

Governance Structure:

The governance model is structured into **three key entities**:

- **General Membership (HRT Holders)**
  - Any Amnesty member can hold **Human Rights Tokens (HRTs)** to participate in governance.
  - Members can comment on proposals, vote, and stake tokens for funding decisions.
- **Delegated Human Rights Representatives (DReps)**
  - Trusted representatives elected by members to act on governance matters.
  - Responsible for protocol updates, treasury management, and partnership approvals.
- **Treasury Council (Multi-signature & Smart Contracts)**
  - Manages the treasury, ensuring funds are allocated to impactful projects.
  - Raising proposals, ethical standards, legal and regulatory.
  - Uses a mix of **quadratic funding + impact verification** to distribute resources.

Decision-Making Process

- **Proposal Submission**: Any HRT holder can vote on governance proposals.
- **Community Deliberation**: Discussions and feedback loops are held on-chain.
- **Voting Mechanism**: Members vote using either **token-based voting (HRTs) or quadratic funding models** to ensure fair participation.
- **Execution via Smart Contracts**: Approved decisions are implemented through **Cardano smart contracts**.

To manage the increasing complexity of governance proposals, collaborations with the Cardano Foundation have led to the development of AI agents aimed at streamlining the evaluation process. These agents include:

- **DRep Aid**: Assists DReps by suggesting votes on proposals based on predetermined criteria.
- **Sentiment Analysis**: Captures and analyses community sentiment on governance proposals.
- **News and Context Analyser**: Provides relevant news and context for governance proposals.
- **Autonomous DRep Agent**: Acts as a fully autonomous DRep, voting on proposals.

These tools aim to enhance transparency, scalability, and community empowerment in Cardano's governance processes, and will no doubt be enhanced by rapid ongoing iteration in this area.

2 Enhanced Voting Mechanisms and Reputation-Based Governance

Strategy: Adopt a hybrid voting approach combining the strengths of multiple systems while mitigating their weaknesses.

Preventing Token-Based Plutocracy

One of the biggest risks in DAO governance is plutocracy—where those with the most tokens control decision-making. This is especially problematic in human rights advocacy, where financial influence should not outweigh real activism and expertise. HRDAO solves this problem through reputation-based governance.

How Reputation-Based Voting Works

Reputation-based governance is where voting power is based on a member's contributions or reputation, rather than the number of tokens they hold or use to vote. This model encourages merit-based participation.
Instead of pure token-based voting, HRDAO will integrate this reputation-weighted system that considers:
Activist Contributions—Verified human rights advocacy work increases reputation.
DAO Engagement—Participation in discussions, voting, and proposal creation earns reputation points.
Funding Impact—Proposals that create measurable human rights improvements boost a user's reputation.
Skills Training—verified upskilling and capacity building of a participant and extending to the network enhances governance rights.

We will need to ensure reputation systems are transparent, contestable, and not gamified (social credit scoring risk). Reputation would need to come with a Reputation Ethics clause — e.g., scoring must:

- Be contextual (geography, threat level)
- Be revocable
- Be protected by privacy layers

  Reputation is verified trust, built through care, action, and presence. And like trust, it must be accountable and protected.

Quadratic Voting and Delegated Representatives (DReps)

To further balance governance participation and influence, HRDAO employs quadratic voting (QV) and Delegated Representatives (DReps). **Quadratic voting** addresses the problem of token whales having disproportionate power by making each additional vote cost more tokens. This model ensures that while large holders or influential individuals still have a voice, the collective power of smaller token holders is not overshadowed, as recommended by Shah (2024). This mathematical approach reduces the outsized impact of "whales" while still respecting the principle that those with more at stake should have meaningful influence. For long-term strategic decisions, conviction voting can also reward commitment and discourage last-minute manipulation, addressing concerns about governance token manipulation raised by Kitzler et al. (2023).
Quadratic voting ensures that all voices matter, but prevents wealthy actors from dominating decision-making. Instead of one-token-one-vote (which enables plutocracy), QV makes each additional vote exponentially more expensive, ensuring fairer governance. In this way, a small group cannot monopolise decisions, minority viewpoints are protected and it encourages consensus-driven governance.

Example: If a user wants to cast:

- **1 vote** → costs **1 unit of voting power**
- **4 votes** → costs **16 units**
- **9 votes** → costs **81 units**

This prevents the wealthy from disproportionately influencing outcomes while still allowing meaningful participation.
It is clear with **10 million+ Amnesty members worldwide**, the DAO must **prioritise grassroots participation**.

Delegated Representatives (DReps)

Delegated governance allows token holders to delegate their voting power to trusted representatives called DReps. This helps streamline decision-making by concentrating votes amongst informed delegates, and ensuring those who are time poor can still have an influence on outcomes. DReps act as trusted intermediaries for users who cannot participate in every governance vote but still want a voice. They:
represent large groups of users in decision-making,
are elected based on reputation and governance participation, and
ensure broad representation of diverse Amnesty communities.

DReps align with Cardano's SIP-1694 model, integrating HRDAO into the broader Cardano governance framework.

Key Governance Tools Summary

	Function	**Why It's Important**
Smart Contracts	Automate governance rules	Prevents corruption, ensures fairness
Quadratic Voting Protocol	Balances voting power	Prevents plutocracy, enhances

	Function	Why It's Important
		democracy
Reputation Tracking System	Assigns governance power based on activism and roles	Rewards real contributions
On-Chain Proposal System	Enables structured governance proposals	Ensures transparency and accountability

To further diversify decision-making, we could incorporate **Sortition** (random selection) for forming specialised committees, ensuring representation beyond the usual power players. Research by Ballandies et al. (2024) emphasises that DAOs are complex systems requiring bottom-up self-organisation, which these varied approaches can facilitate. Each voting mechanism can be tailored to the specific decision context—using simpler mechanisms for routine matters and more sophisticated approaches for high-stakes decisions.

Random sortition introduces a system where governance committee members are periodically rotated based on algorithmic selection. This reduces the possibility of bias, the potential to pre-select candidates and brings in a variety of potential viewpoints, leading to more comprehensive and inclusive decision-making.

As the DAO scales, active participation will be key to keeping the DAO community-driven (and thereby more functionally decentralised). Scale will also bring in considerations of AI automation assistance to streamline proposal management and administrative workload, and the possible implementation of staking, vesting schedules or rewards to incentivise long-term participation and commitment, further to reputation-based non-transferable tokens. There is a number of potential voting mechanisms that could be built into the future DAO vote-weighted matrix as the platform engagement grows and voting nuance matures. [14]

This hybrid approach acknowledges there is no perfect voting system, but rather different systems that excel in different contexts. By thoughtfully applying the right mechanism to each decision type, the DAO can navigate the trade-offs between participation, expertise, decentralisation, and efficiency that Sharma et al. (2024) identified as critical governance considerations.

3 Participation Incentives and Requirements

Strategy: Create a comprehensive system of incentives and requirements that encourage meaningful participation and consensus-building. The aim is Regenerative Coordination.

We can intuit that consensus goes beyond a simple majority but stops short of requiring total unanimity. In this sense, it compels a degree of collaboration—those who may not fully agree with the majority decision nonetheless accept being bound by it. As Edward Shils explains, consensus *"reinforces the cooperation which arises from coincidences of interest [and] limits the range of divergence by defining ends in a way which renders them more compatible."* [15]

Achieving consensus not only strengthens collective decision-making but also fosters a renewed focus on core missions and the broader human rights impact we seek to achieve. If we focus on experimentation, a potential diversity of inputs and feedback loops, we can centre a principle termed "maximal interestingness" – the idea that the most interesting ideas naturally gain momentum and support – especially within a system aligned on shared purpose (in this instance the protection and celebration of human rights).

We can also consider the concept of "**rough consensus**" – whereby the objective is to find the most fertile direction to head in or decision to make, whilst resisting limiting visions. This is balanced with the obligation to clearly hear substantive objections, but not allow them to inhibit progressing in alignment with the overall vision.

To support this concept in the HRDAO, participants must be incentivised to engage meaningfully in the decision-making process. Additional to the reputation weighted process outlined above, this can include rewards for active participation, such as voting rights, recognition, or access to decision-making opportunities, while also establishing clear requirements for involvement—such as a commitment to abide by the collective agreements. These incentives and requirements ensure that all participants have a shared stake in the outcomes, encouraging alignment with the group's goals and reinforcing the collaborative nature of consensus-building.

The HRDAO should frame consensus as both a collective responsibility and a way to ensure that participants are motivated to align their actions with the broader mission.

For successful implementation, we need to address the participation crisis documented by Falk et al. (2024) by designing a tiered incentive structure that rewards participants for their active involvement in the decision-making process. One option is to assign **reputation scores** based on engagement, such as voting frequency, participation in discussions, and proposal contributions. Higher reputation can grant additional privileges, such as increased voting power or access to leadership roles within the DAO, ensuring that those who contribute meaningfully are empowered to steer the group's direction.

Incentives can also use **tokens or other rewards** to recognise contributions, ensuring participants are rewarded for their commitment. Additionally, clear participation requirements should be outlined in the DAO's governance framework, mandating that participants adhere to collective decisions once consensus is reached. This may include stipulating that participants commit to support final decisions, even if they initially disagree, thereby reinforcing a sense of shared responsibility.

This approach encourages **alignment with the broader human rights mission** by creating a system where participation is valued and incentivised. It ensures that all members have a stake in the success of the initiative, fostering collaboration while maintaining accountability and reinforcing the collective effort to achieve the group's core objectives.

For those less motivated by these incentives, we can develop non-financial rewards like badges, special access, or recognition that appeal to intrinsic motivation and community belonging works. We can also ensure the ability to earn tokens and have your say at an entry level is simplified and democratised, so that the utility to unlock everyday aspects of human energy tied to social good and intention are valued. Laturnus (2023) found that voting activism, rather than token distribution, is a key driver of success, suggesting that cultivating an engaged culture is more important than technical mechanisms alone. This is a crucial point, and one that has driven the Amnesty global movement itself over the years.

We should also consider implementing minimum participation thresholds for certain decisions to ensure legitimacy, with adaptive quorum requirements based on the importance and impact of proposals. This addresses the low participation concerns highlighted by Sharma et al. (2024) while ensuring that critical decisions have sufficient community backing.

The goal is to cultivate a **dynamic feedback loop** where participation becomes ingrained in the community's culture, evolving from external rewards to intrinsic motivation as the community matures. By crafting incentives that cater to diverse motivations, the DAO fosters a robust, self-sustaining participation ecosystem. This mirrors Ostrom's concept of trust-building in the commons, where

"It is not only that individuals adopt norms but also that the structure of the situation generates sufficient information about the likely behavior of others to be trustworthy reciprocators who will bear their share of the costs of overcoming a dilemma." [16]

Ultimately, while code can function as law, it is trust in community consensus—and having skin in the game—that defines true governance. Users must place trust not only in the technology itself but also in the integrity of the network and its governance architecture. Human oversight and energy remain crucial.

4. Transparency Framework

Strategy: Establish robust transparency mechanisms that make governance processes accessible and understandable.

We respond to the documentation deficiencies identified by Ma et al. (2024) by requiring standardised proposal documentation with clear objectives, expected outcomes, and impact assessments. As well as creating templates and guidelines that make it easy for proposers to provide comprehensive information while ensuring that voters can easily compare proposals on key dimensions.

As this project develops, we will create a Decision Matrix Map showing:

- What gets decided by whom
- What threshold is required
- What voting system is optimal

We will implement on-chain voting for all significant decisions, as recommended by Ma et al. (2024) and Tsing et al. (2025), ensuring that voting processes are transparent and immutable. This addresses concerns about vote manipulation raised by Kitzler et al. (2023) by creating a verifiable record of all governance activities.

There is a need to develop visualisation tools and dashboards that help members understand voting patterns, power distribution, and governance trends within the DAO. De Filippi et al. (2020) noted that blockchain increases confidence through cryptographic rules rather than eliminating trust entirely—these tools translate complex blockchain data into intuitive visuals that build this confidence.

Further, we aim to establish regular governance reporting that summarises decision outcomes, participation rates, and governance health metrics, making the state of the DAO accessible even to members who don't follow every proposal. Dispute resolution processes should be transparent possibly using multi-sig councils or off-chain arbitration.

This comprehensive transparency approach transforms governance from an opaque process into a clear system that members can trust and verify, addressing the concerns about manipulation and centralisation documented across multiple studies.

5. Education and Onboarding

Strategy: Develop a comprehensive education program to help members understand governance processes. Embed translation functionality that is locally sourced and decentralised.

We can combat the low participation identified by Falk et al. (2024) by creating interactive tutorials that explain voting mechanisms, proposal processes, and governance philosophy. We will need to make these resources engaging and accessible, using different formats to accommodate various learning styles and knowledge levels.

Part of the community education and onboarding process is to establish mentorship programs that pair experienced and new members, fostering personal connections that increase engagement while transferring governance knowledge. This addresses the knowledge asymmetries that Ma et al. (2024) identified as barriers to broad participation, while building community bonds that encourage continued involvement.

Also there is a need to develop progressive governance participation paths that gradually introduce members to more complex aspects of governance as they gain experience. Beginning with simple voting on well-explained proposals, advancing toward proposal creation, committee participation, and eventually leadership roles. This graduated approach builds confidence and competence simultaneously, creating a pipeline of engaged community members.

Laturnus (2023) found that high voting participation correlates with stronger DAO performance, making education not just a nice-to-have but a core driver of organisational success. By investing in members' governance capabilities, the DAO builds a more resilient and sustainable decision-making ecosystem that can adapt to changing conditions and address the complex challenges identified in the research.

Here we could incorporate a **Proposal Ethics Checklist**, where each submission must include:

- Human rights impact
- Accessibility
- Risk analysis
- Ethical alignment

On Incorporating Multilingualism and Decentralised Translation

Language is not just a tool of communication — it is a vessel of worldview, resistance, and belonging. In building the Human Rights DAO (HRDAO), we must recognise and build for this. The pluriverse of human experience demands that our governance and participation layers are not limited by linguistic imperialism or centralised translation engines. To truly empower a global movement, we must build a decentralised language infrastructure that reflects and respects local knowledge systems.

Practically, this means embedding translation functionality directly into the DAO's tooling — not as a monolithic API from a Silicon Valley provider, but as a modular and community-curated translation layer. Leveraging decentralised protocols such as InterPlanetary File System (IPFS), Filecoin or Arweave, language packs and glossaries can be locally hosted and dynamically updated by contributors in their native tongues. A reputation-based system, backed by token incentives or soulbound trust scores, can ensure high quality and contextually appropriate translations — including those for activist-specific language, legal terminology, or indigenous epistemologies. An interesting challenge would be to include Indigenous language protocol compatibility (e.g., writing right-to-left, visual symbols).

We affirm that multilingual participation is a prerequisite for a truly equitable and global digital governance system. Language here becomes a governance primitive.

6. Ethical Safeguards

Strategy: Embed ethical principles into the core of governance design.

"several blockchain communities have begun adopting off-chain governance mechanisms that are substantive in nature, most notably drafting constitutions and code of ethics to articulate shared values that they agree to be bound by..." [17]

Following the recommendation of Han et al. (2024) we aim to develop a community-approved ethical constitution that guides decision-making. This document should articulate core values, prohibited actions, and ethical boundaries that govern the DAO's operations, providing a reference point for evaluating all governance decisions.

Also we aim to implement specialised committees or councils focused on ethical oversight, both within Amnesty leadership and with external stakeholder group participation, with the authority to review proposals for alignment with community values. This addresses Kitzler et al.'s (2023) finding that ethical principles should be hardcoded into governance design to prevent malicious actors from shifting DAO priorities.

There is also a need to create anonymous reporting channels for ethical concerns with defined resolution processes, enabling whistleblowers to flag potential issues without fear of retaliation. This responds to the governance token manipulation risks identified by Kitzler et al. (2023) by creating accountability mechanisms that don't rely solely on token-based voting.

Regular ethical audits can evaluate governance decisions against the community's stated values, ensuring alignment between actions and principles over time. By treating ethics as a fundamental design constraint rather than an afterthought, the DAO can build governance systems that serve the community's true interests rather than merely following procedural rules. This ethical foundation is particularly important given the findings of Manzano, Kharman & Smyth (2024) that economic incentives can encourage behaviours that undermine fair governance.

To ensure governance aligns with Amnesty's mission, core ethical values are hardcoded into governance rules, making them resistant to malicious amendments.

Key Features:

- **Immutable Amnesty DAO Charter:** The DAO's **foundational principles** (e.g., commitment to human rights) can only be changed by a supermajority.

- **Human Rights Impact Assessment (HRIA) for Proposals:** Every proposal must include a documented assessment of its potential **impact on human rights**.
- **Emergency Veto Mechanism:** A multi-stakeholder ethics panel can **pause proposals** that violate human rights principles.

Structural Checks and Balances:

To prevent governance manipulation, HRDAO implements **transparent decision-making layers**:

- **Ethical Guardians Council:** A randomly selected **council of human rights experts** reviews and advises on governance decisions.
- **Liquid Democracy Mechanism:** Members can **delegate votes to trusted representatives** for specific topics.
- **Community Consensus Protocol:** Major governance shifts require both **token-holder approval and deliberative human rights review.**

Transparency & Accountability Mechanisms:

To protect against corruption and power centralisation, Amnesty DAO ensures full transparency and **real-time auditing of governance decisions.**

- **On-Chain Voting & Public Records:** Every vote and decision is recorded on-chain, ensuring **public scrutiny**.
- **Whistleblower & Protection Protocols:** Anonymous reporting channels allow members to **flag governance abuses**.
- **Voting Behaviour Audits:** Periodic audits of voting patterns ensure no single entity is manipulating outcomes.

Further to the above governance-based ethical issues, we need to also touch on how aspects of the DAO can more generally support freedom of engagement (and therefore more robust governance), and human rights protections.
The HRDAO will support fundamental freedom of expression by affording the pseudo-anonymity of its participants, and the documentation and engagement of refugees, the stateless and unbanked. The technology

"could also enable personal autonomy by facilitating information self-determination," [18]

and freedom of expression for participants, and also ensure engagement is not explicitly linked to these same participants. Key to this ability to enhance privacy and therefore safety in a human rights context for participants in the novel technology called zK proofs (zero-knowledge proofs). **zK proofs** are cryptographic methods that allow one party to prove to another that a statement is true without revealing any information beyond the truth of the statement itself—balancing privacy and verification. We will incorporate this technology through the Midnight blockchain's functionality.
Its important to note also that as the technology matures and engagement in the platform develops, its applications for resolving human rights issues will expand.

Blockchain technologies also introduce novel ways of encoding and enforcing human rights. Blockchains and DAOs specifically, according to Nathan Schneider:

"have the potential to create a new layer of global social contracts, in which human peers, more

than territorial governments, are the protagonists."
He goes on:
"What I am suggesting is more foundational than what people usually think of when they discuss blockchain for human rights: new apps for tracking supply chains or verifying votes or protecting news reports from censorship. Blockchains themselves could have human rights written into their basic protocols... [and] have the potential to create a new layer of global social contracts, in which human peers, more than territorial governments, are the protagonists" [19]

We have the opportunity to enhance human rights in the digital sphere by centering inclusion, plurality of voices and identities, and by encoding human rights protections in the technical and governance architecture. For, otherwise
"to be neutral on human rights is in fact a choice not to consider human rights. Neutrality is an implied refusal, a missed opportunity, a failure of imagination." [20]

In fact, one of the first tasks of a fully operational human rights DAO would be to define exactly what those rights look like on the platform.

7. Adaptive Governance Framework

Strategy: Build flexibility into the governance system to allow for evolution and improvement.

We can address the complex systems nature of DAOs identified by Ballandies et al. (2024) by scheduling regular governance reviews with community input. These structured evaluations can assess what's working, what's not, and what needs to change, creating formal opportunities to adapt governance processes to changing circumstances.

As a result we can establish clear amendment processes for governance rules, with different requirements based on the fundamental nature of the change. Minor operational adjustments might require simpler approval processes, while constitutional changes could demand supermajorities or extended voting periods to ensure broad consensus.

By using graduated changes rather than sudden overhauls to maintain stability while allowing innovation, we recognise the declining decentralisation in DeFi and investment DAOs over time observed by Sharma et al. (2024), while creating mechanisms to detect and correct governance drift before it becomes entrenched.

We will need to test governance experiments in controlled environments before full implementation, allowing the community to test new approaches with limited risk. This evidence-based approach to governance evolution ensures that changes are driven by results rather than ideology, addressing the empirical findings across studies that many governance innovations have unexpected consequences when implemented at scale.

The ability to adapt is perhaps the most important meta-governance capability, as it allows the DAO to respond to new research insights, changing external conditions, and evolving community needs over time. Constant iteration and feedback are a crucial aspects when building on the initial DAO MVP and testing phase.

Introduce tiers of engagement:
Anonymous → Credentialed → Representational to reflect different risk levels of DAO participation.

To ensure the HRDAO reflects real-world legitimacy and cultural depth, we again highlight a proposed *Embodied Governance* layer — a decentralised cultural-ethical framework that honours consent, legitimacy, and inclusion beyond code. This layer foregrounds storytelling, care, and relational consent as foundational to participation, particularly for marginalised and at-risk communities. Key components include: culturally adaptive consent protocols (e.g. Ubuntu, adat, or elder validation models), decentralised onboarding rituals such as story circles and narrative-based reputation, and guardianship mechanisms to protect those in high-risk zones through anonymity, delegation, seasonal governance summits and zero-knowledge co-governance. This approach centres dignity, cultural safety, and memory, and recognises that governance, at its core, is relational. As such, HRDAO becomes not only a technological system, but a shared space of trust, story, and solidarity.

8. Community Feedback Mechanisms

Strategy: Establish regular feedback loops with stakeholders to ensure the DAO evolves according to community needs and expectations. We will establish a creative problem solving and feedback process based on humility, equity and multi-voiced co-design.

Feedback Mechanisms:

Monthly Stakeholder Meetings: we link with Amnesty tech leadership and regional stakeholders to ensure the project is aligned with the latest advocacy and campaigning objectives. We gather feedback and allow for open dialogue, enabling participants to voice their opinions and ask questions.

Sector Advisory sessions: we will establish a team comprised of diverse community representatives who can provide insights and feedback on project initiatives and governance. These experts include active members of the Cardano and Project Catalyst community as well as other aligned stakeholders in the Midnight Blockchain, UNDP, the Blockchain for Good Alliance and other impact-focused DAO projects. We include members from different geographic regions, backgrounds, and advocacy areas.

Public MVP Testing: the HRDAO MVP needs real-world testing among a diverse subset of Amnesty members before full roll-out. The plan is to test the DAO MVP in Q3-4 2025 with Amnesty stakeholders and specifically targeting the advocacy groups and members in the following participating countries: New Zealand, Australia, Taiwan and Thailand.
Some of these advocates are in high risk zones, so have enhanced privacy requirements.

Testing Governance Mechanisms:

- We will run simulations of earning and burning the HRT-UTL token, to see:

  - if users understand and can effectively participate in these mechanisms.

  - the ability to conduct low-stakes tokenised votes on sample campaigns (e.g., proposed Amnesty advocacy initiatives).

  - to test adaptive quorums to estimate realistic thresholds for participation rates.

Social Infrastructure:

this refers to the ongoing support and communication with aligned institutions, and the wider advisory network within Web3 that we are operating in. We aim for deeply relational practices of engagement, with an understanding that power relations can be transformed through the influence of diverse actors, and the activation of these "social layers".
These crucial Web3 community connections will guide and assist the project as it proceeds through testing and building phases. Within this process we are aware of integrating deep listening – to uncover unconscious bias, raise cultural awareness and emerging power dynamics, question narratives and unlock imagination, ideas and potential. In this way we are "designing disorder" – enhancing collective decision-making, fluid exchanges, and making space for participation and inclusion. We aim to further develop and entrench our culture of belonging and connection in Amnesty, centered on our human rights purpose.

Quantitative and Qualitative Metrics – full DAO model:
For the fully realised DAO model we would want to determine various metrics for engagement.

Quantitative Metrics

Voting Participation Rate:

- **Definition**: The percentage of community members who participate in governance votes. Also monitor the percentage of feedback generated from Global South regions or marginalized groups.
- **Target**: Aim for at least 30% participation in each voting cycle.

Proposal Submission Rate:

- **Definition**: The number of proposals submitted by community members within a specific timeframe.
- **Target**: Track an increase in submissions each quarter.

Community Engagement Metrics:

- **Definition**: Measure the number of active users on community platforms (forums, Discord, etc.).
- **Target**: Set a goal for growth in active users by 20% each quarter.

Funding Allocation Transparency:

- **Definition**: Track the percentage of funding allocations that are publicly reported and accessible to community members.
- **Target**: Aim for 100% transparency in funding decisions.

Impact on Advocacy Campaigns:

- **Definition**: Number of advocacy campaigns initiated through DAO support and their measurable outcomes (e.g., petitions signed, media coverage).
- **Target**: Report on campaign success rates annually.

Feedback Response Time:

- **Definition**: Average time taken to respond to or implement significant feedback
- **Target**: Report on campaign success rates annually.

Qualitative Metrics

Community Satisfaction Surveys:

- **Definition**: Analyse feedback from surveys focused on member satisfaction with the DAO's processes and governance.
- **Goal**: Achieve an average satisfaction rating of 4 out of 5.

Sentiment Analysis:

- **Definition**: Conduct sentiment analysis on feedback collected from community discussions, social media, and forums.
- **Goal**: Aim for a sentiment score indicating positive engagement (e.g., >70% positive feedback).

Case Studies of Successful Initiatives:

- **Definition**: Document qualitative stories of successful initiatives and their impact on human rights advocacy as a result of the DAO's support.
- **Goal**: Compile and publish at least 3 case studies annually.

Diversity of Voices in Decision-Making:

- **Definition**: Assess the representation of marginalized voices in proposals and decision-making processes.
- **Goal**: Ensure at least 50% of proposals come from underrepresented groups.

Feedback Quality Assessment:

- **Definition**: Evaluate the quality and constructiveness of feedback received through community channels.
- **Goal**: Aim for at least 75% of feedback to be actionable and constructive.

<u>Conclusion:</u>

Amnesty HRDAO is not just another governance experiment—it is a **blueprint for ethical, decentralised governance in the human rights and wider NGO sectors worldwide**.
By integrating quadratic voting, incentive tokenomics, tiered governance, transparency, education and structured ethical safeguards, it ensures inclusive, transparent, and resilient decision-making. This governance model aims to **set a precedent for mission-driven DAOs** on the Cardano blockchain, where justice and decentralisation coexist.

Based on the above, we are now able to answer more of our research questions:

How do various models address power imbalances and ensure marginalised voices are included in decision-making?

The HRDAO explicitly challenges traditional token-weighted governance models that often entrench power in the hands of wealthy stakeholders. Instead, it adopts a multi-tiered, hybrid governance architecture that elevates marginalised voices through reputation-based voting and Delegated Representatives (DReps). These DReps are selected not by wealth but by proven advocacy and participation, allowing grassroots activists from the Global South or restricted regions to steer decisions meaningfully. Furthermore, mechanisms like incentivisation and soulbound reputation tokens ensure that governance is not only decentralised but deeply pluralistic—designed to amplify voices historically excluded from institutional processes. This is decentralisation with a conscience, not merely a technical architecture.
HRDAO prioritises activism over token holdings using reputation-based governance and quadratic voting (QV) to prevent dominance by large token holders. Additionally, Delegated Representatives

(DReps) enhance decision-making by bringing expertise-based governance. These mechanisms ensure that governance is inclusive, expertise-driven, and mission-aligned.

What metrics or indicators are most effective for measuring the impact of governance models on social justice or advocacy outcomes?

HRDAO proposes a set of precise, on-chain metrics to track impact in ways that reflect both ethical alignment and functional output. These include: *Proposal Inclusivity*—how many funded proposals originate from marginalised groups; *Funding Transparency*—measured via smart contract-based treasury tracking; *Voting Participation*—counting unique voter engagement per cycle; and *Advocacy Impact*—assessed through traceable real-world policy or campaign outcomes. These metrics are instruments of collective accountability. By aligning data tracking with mission delivery, HRDAO transforms governance into a mechanism of radical transparency and activist accountability. Successful decentralised governance models share three essential qualities: accountability mechanisms, transparent voting, and equitable participation. These principles ensure governance remains trustworthy, inclusive, and resistant to corruption. HRDAO integrates these best practices through reputation-based and quadratic voting models, ensuring that human rights activists have a meaningful voice in governance rather than being overshadowed by wealth-based influence.

It is planned for the fully realised HRDAO to eventually employ on-chain analytics to measure its social justice impact. Key metrics include proposal inclusivity, tracking the number of proposals from marginalised groups that receive funding; funding transparency, monitored through smart contract-based treasury allocation; voting participation, measuring unique voter engagement per governance cycle; and advocacy impact, assessing measurable policy changes achieved.

Which governance mechanisms are most appropriate for scaling human rights-focused initiatives?

Quadratic voting and reputation systems form the ethical core of the HRDAO governance stack. Quadratic voting mitigates plutocracy by increasing the cost of each additional vote, ensuring no single actor can dominate outcomes. When paired with non-transferable, soulbound governance tokens tied to real-world activism, this system privileges lived experience and commitment over capital. Token-weighted voting is deliberately de-emphasised except in low-stakes campaign-level decisions. Instead, governance rights are earned through contribution, peer endorsement, and verified advocacy—ensuring that the DAO scales with integrity and inclusivity. It's governance by trust, not transaction.

By learning from past governance failures (e.g., The DAO Hack, Aragon DAO struggles), HRDAO prioritises security through audited contracts, active governance participation via DReps, and anti-plutocracy safeguards through QV and reputation-based voting. HRDAO establishes a robust, scalable, and equitable governance framework that enhances human rights activism through decentralised technology.

7 Tokenomics Modelling

Tokens are defined as units of value created by organisations to facilitate interactions within their ecosystems, in this case the Amnesty HRDAO. They represent various forms of value, including currency, utility, or ownership.
Tokenomics defines the economic model of the HRDAO, affects the project's long-term viability and growth and market value, and drives user behaviour and engagement.

The foundation of any successful project is a **clear token value proposition**, strategy, and intended user base. Having a clear idea of the various stakeholder groups, their preferences and outside options, and their constraints lays a foundation for a successful economic design. All the economic design in the world is of limited use if—at the end of the day—the platform does not deliver value to its various stakeholders. As Schneider (2022) comments

"Having analyzed over 100 token systems, it seems that the clearer the purpose, the more resilient the network." [21]

The idea of utility gives the token organic demand, which in turn drives value and liquidity. For example, by offering solutions to problems, having a loyal and engaging community, and other innovative features, a project might see an increase in its demand.
Trust plays an essential role in the utility of tokens. When tokens represent the right to access a service or participate in a vote within a regulated ecosystem, the token holder grants trust to the token issuer.

Tokenomics: Designing Value with Purpose

Tokenisation is a mechanism for networks and communities to incentivise collective value creation and enable contributors to partake and share in the value captured.

At the heart of the HRDAO lies a simple yet profound principle: the value of a token must be anchored in trust, purpose, and community. In this ecosystem, tokens are not just digital assets—they are instruments of agency, tools of governance, and keys to participation. They must carry the ethical weight of the Amnesty movement they serve.

The HRDAO proposes a **DUAL-TOKEN Model**—one that balances functionality with philosophy:

1. **Utility Token (HRT-UTL)**: Fungible, transferable; used for campaign voting, staking, micro-rewards, and operational access.

These are the living currency of the HRDAO, enabling access, action, and alignment. These tokens embody participation—they track micro-contributions, reward effort, and unlock tools that empower human rights defenders globally. Utility is essential—without use, value is unrealised.
Key functions include –
Access to Tools – unlock digital services and human rights platforms;
Work Rewards – earn tokens through advocacy, research, and contribution;
Staking for Participation – represent commitment and presence within the DAO;
Ecosystem Exchange – facilitate peer-to-peer value transfers in a trust-minimised system;
Engagement Mechanisms – build "skin-in-the-game" relationships that drive deep involvement;
Proposal Voting Rights – participate in decision-making on proposals initiated by Governance token holders.

Utility tokens are the breath and bones of our collective architecture—animating the DAO through everyday use and meaningful action.

2 **Governance Token (HRT-GOV)**: Soulbound, non-transferable; earned through verified contributions and reputation; required for protocol-level proposals and constitutional changes.

These tokens are soulbound / non-transferable reputation tokens. These grant decision-making power, allowing members to shape the DAO's direction and steward its mission. Governance tokens are designed to support high participation and meaningful stewardship.
Key features include:
Decentralised Voting – token-weighted or quadratic;
Progressive Delegation – allowing trusted voices to emerge;
Proposal Thresholds – to ensure thoughtful governance;
Staking for Voice – holders can stake tokens to gain governance influence;
Reputation-based Influence – long-term contributors accrue credibility, amplifying their voice and impact over time.

If users have the perception that the platform operates in good faith and according to human rights standards, then governance becomes deeply embedded, not just functional.

Complexity vs Elegance: Why Two Tokens?

The dual-token model, gives each element a distinct and intentional role. Utility tokens function like a digital "key" to shared infrastructure, allowing a voice and input into campaigns and actions. Governance tokens, by contrast, bestow the power to raise proposals to be voted on, debate, and vote on decisions that shape the future of the actual DAO. This separation is not just functional—it is philosophical. It allows economic clarity while honouring the sacred responsibility of reputational governance with its own dedicated token form. It also protects against undue speculation distorting governance processes. We would term these governance tokens as "soulbound" – meaning non-transferable, and also non-fungible (meaning each token is unique to the holder and is non-equivalent to each other governance token).
The dual-token model allows clarity. The **Utility Token** powers the daily operations of the DAO—services, work, transactions. The **Governance Token** anchors participation in collective decision-making. Separation of utility from governance enhances clarity, protects token functions from price speculation, and empowers communities to govern in peace. This separation of economic utility and reputational governance speaks to the very heart of pluralistic, trust-based communities. It echoes the wisdom of Elinor Ostrom's commons theory and the Indigenous jurisprudential principle that **different forms of value—material, spiritual, reputational—should not be collapsed into a single transactional logic.** In many Indigenous and customary legal systems, the right to speak is earned through relationship, not purchase. The HRDAO seeks to encode this principle—to ensure that the voice of the community is not drowned out by liquidity, but amplified through protocols of inclusion, custodianship, and care. This dual-token model allows us to model a more ethical, post-extractive web3 architecture.

By distinguishing between the flow of resources and the practice of governance, we safeguard the integrity of each domain. The Utility Token honours labour, reciprocity, and functional contribution. The Governance Token honours presence, wisdom, and responsibility in collective care. This reflects a deeper epistemology—one where participation is not simply a function of stake, but of standing; where governance is not dominated by capital, but guided by consent, trust and a commitment to equitable shared futures. However, we need to ensure user confusion is at a minimum and mitigate any unclear UX/UI and education gaps with layered onboarding that allows people to *gradually* move from Utility to Governance roles. We are also considering token lifecycle and redemption mechanics (especially for the Utility Token). What happens if Utility tokens are hoarded, lost, or exploited by spammy participation farming? We need to anticipate mitigation mechanisms and respond to real world tokenomic vectors. Here we could embed a **"rite of passage"** mechanic: HRT-UTL users must complete a set of campaigns, earn endorsements, or pass peer review before accessing GOV eligibility; and make use of interactive narratives or journeys (e.g., "Path of the Activist" or "Guardian Journey") to shift from transactional engagement to meaningful presence.

The **Utility Token** in the HRDAO ecosystem is designed first and foremost as a medium of meaningful exchange—a recognition of contribution, service, and relational labour within the community. In its initial phase, it functions as a non-speculative unit of value that rewards participation, builds trust networks, and fuels the daily operations of the DAO. The early architecture emphasises mutual recognition, reciprocity, and honouring the time, skills, and care invested by members. This aligns with the idea that value is first social before it becomes financial —a principle found in both Indigenous economies and regenerative economic models.

Over time, and as the ecosystem matures, the Utility Token may evolve to interact with broader systems of exchange. This could include mechanisms for stable value representation or **selective monetisation**, especially in contexts where it becomes important to bridge internal reputation economies with external currencies. Such a shift would only be explored in ways that preserve the integrity of the DAO's founding purpose and aligned with the broader Amnesty human rights mission.

In this light, the Utility Token is not simply a transactional tool—it is a seed of future value, whose worth will be determined by the strength of the community, the richness of its interactions, and the ethical clarity of its design. Whether or not it later integrates with financial systems, it will always retain its primary purpose, and its primary value: to circulate trust, enable action, and honour participation within the HRDAO commons.

Summary:

Token Type	Role	Transferable	Cardano Tooling
HRT-UTL	Utility, Operations, Work	Yes	Native asset, no contract needed
HRT-GOV	Governance, Soulbound, Trust	No	NFT + Plutus policy, CIP-68, optional DID + zK

HRDAO Dual Token Model — Clarified:

1. Utility Token – HRT-UTL (Human Rights Token – Utility)

Nature: Fungible, transferable
Held by: Anyone participating in day-to-day DAO actions

- Used by activists, members, builders and organisers—anyone interacting with the DAO's operational layer.
- Used by journalists, artists and lived experienced to verify and crowdfund activity
- Amplified and exchanged by participants to signal preference and support
- Native Asset using Cardano's multi-asset ledger. Does not require smart contract to mint and transact.

Minting Tools:

- Mint on-chain using tools like Cardano CLI, or UI platforms such as TokenForge, FluidToken or MeshJS.

Optional: Implement staking mechanisms using Plutus contracts.

Earned by:

- Participation in micro-actions, local activism or artivism
- Campaign tasks, campaign focus voting, content creation
- Learning and education modules
- Referrals, community organising

Core Functions:

- Access to DAO services and tools (communication, action platforms, knowledge banks)
- Used for incentives, gas, and transaction-like operations
- Vote on proposals raised by Governance Token holders—participatory democracy
- Weighted voting may be quadratic or 1:1, based on contribution

Voting Rights:

- YES: Campaign-Level Voting (e.g. "Should we launch this campaign on climate justice?")
- NO: Protocol or constitutional proposals (reserved for HRT-GOV holders).

2. Governance Token – HRT-GOV (Soulbound Reputation Token)

Nature: Non-transferable, soulbound (1 per identity)
Held by: Proven contributors, Amnesty stewards, verified advocates

Purpose:

- Participatory governance: proposal-making, and community accountability.
- Signifies reputation, trust, and standing within the HRDAO—non-fungible and non-transferable by design.
- Soulbound Token (non-transferable NFT) or non-fungible native asset with minting policy that disallows transfers.

How to implement Soulbound logic:

- Write a Plutus policy script that only allows:
    - Minting (once) by the DAO.
    - Burning (by DAO or under condition).
    - No transfers—token is locked to a specific address.
- Could be issued upon proven contribution or through quadratic voting models.
- Can integrate reputation metrics (e.g. voting history, campaign support).
- Tools to use include Tally or Snapshot for any off-chain voting, Liqwid for reputation-weighted staking.
- **CIP-68 (NFT metadata standard)** – crucial if governance tokens need embedded logic or metadata.

Earned by:

- Demonstrated activism, consistent support, campaign success
- Passing threshold of learning/knowledge missions
- Community endorsement, peer-review or steward nomination
- Leadership roles existing with Amnesty and verified stakeholder/partner groups

Core Functions:

- Right to create proposals (protocol-level or campaign-level)
- Right to vote on protocol governance: treasury policy, tokenomics, system upgrades
- Ability to mentor or endorse new contributors, right to vote on increased governance responsibility for participants.
- May serve in multi-sig or guardian councils for constitutional changes

Voting Rights:

- YES: Protocol-Level Voting (e.g. "Should we change the utility token inflation model?")
- YES: Campaign Proposal Creation (e.g. "Here's a 6-month advocacy plan on digital privacy in MENA")
- Can trigger referenda for utility-token holders to vote on.

  Optional Enhancement: DID Binding for Governance Tokens:

  To deepen the identity-linkage and preserve soulbound ethos we could bind governance tokens to DID (Decentralised Identifiers) using Atala PRISM. This enables verifiable credentials: e.g. "This wallet address belongs to an Amnesty-trained community advocate".

The Separation — *Why it Matters*

Role	HRT-UTL (Utility)	HRT-GOV (Governance)
Earned Through	Action & Engagement	Proven Reputation & Deep Advocacy
Transferable?	Yes	No – Soulbound
Use Case	Daily Operations, Participation	Strategic Design, Protocol Safeguarding
Voting Power	Campaign-level voting	Protocol and proposal creation + higher-level votes
Proposal Creation	No	Yes
Philosophical Frame	Access & Contribution	Stewardship, Trust, and Responsibility

Interaction Between Tokens

1. **Governance token holders raise proposals**, design campaign frameworks or vote on structural updates.
2. **Utility token holders** then get to vote on those campaign proposals—enabling broader democratic input.
3. A **feedback loop** allows the most active utility holders to eventually **earn** HRT-GOV, creating a ladder of trust and responsibility.

“The ones who do the work eventually become the ones who shape the work.”

This two-fold complexity must be navigated carefully. Despite the benefits that a two-token model can bring, we also have to note some negatives. Use cases of a token are separated, which can lead to unnecessary complexity. Clear documentation, onboarding education, and UX clarity are essential to avoid alienating participants. We propose this tiered participation architecture that both rewards engagement and protects protocol integrity. This mirrors both **tribal councils and open assemblies**—a logic of care and accountability. We believe this is rare and powerful, and encapsulates the spirit of open access and crowdsourced value, while protecting the community through curated and respected governance.

The Soul of the System: Trust, Utility, and Community:

Tokens must first be **useful** to become valuable. A well-designed token model doesn’t just enable transactions—it fosters meaningful participation and rights-based values into every interaction. A token’s value stems not from market speculation, but from the lived experience of those who use it. When it solves real problems, enhances the user journey, and is stewarded with transparency, demand follows naturally.
Tokens should allow members to access digital human rights tools, reward advocacy work, engage in meaningful exchanges within the ecosystem, or signal participation and reputation. These use-cases are aspects of belonging. Through action and engagement, the token becomes a vessel of solidarity.

And where there is usefulness, there is **trust**. Trust is the architecture beneath all successful decentralised systems. When people exchange tokens for access or stake them in support of governance, they are expressing belief—in the system, themselves and in the mission. Trust creates liquidity, not the other way around. Tokens can initiate multiple layers of utility, encouraging core behaviours and participation in which trust and enrichment of the user experience are key. The aim is to create a flourishing eco-system with a high degree of user engagement. In many respects, this user engagement is already happening through millions of data points across the Amnesty global mission, it just isn't alive and coordinated.

"While the terms of any contract are an essential lever in shaping the incentives of participants, they are but one lever. The founding team must consider how all the many environmental factors come together to shape the economic decision-making of users." [22]

The HRDAO builds trust through:

- Transparent governance processes.
- Clear rules for engagement and participation.
- Systems that reward long-term commitment and crowd-sourced initiatives.

When token holders use their tokens to vote, collaborate, or access services, they are placing belief in the ecosystem. This belief, this trust, is what gives the token its real power, and the **community** its effective voice.

A token's utility may include accessing digital tools, unlocking rights-education content, or recognising contributions from grassroots communities. These functions foster organic demand, as the token becomes a symbol of trust, usefulness, and contribution.

"Demand comes from the token fulfilling a purpose, and being trusted to deliver on this purpose in a sustainable way." [23]

Where purpose leads, value follows.
Tokenomics for human rights and the HRDAO must be regenerative, not extractive; circular, not linear; participatory, not top-down.

As a note we are inspired by both the attention and movement economies and the emerging ideas of value-based metrics in these eco-systems. In the context of blockchain technology and healthcare, incentivising healthy behaviour refers to using transferable tokenised rewards to motivate people to take better care of their health. This approach not only empowers individuals to be healthier, leading to positive health outcomes, but also can potentially reduce the long term health costs of society. Can we do the same for incentivising human rights actions leading to increased *societal* health?

Architecture of a Regenerative Token Economy for the HRT:

Token Supply

Token supply sets the rhythm of a token economy—how tokens are created, distributed, and circulated. This includes:

Will the supply be capped or inflationary? Suggestion is that the supply is in the initial phase inflationary, so the DAO is not to rigid in responding to new user growth, and network needs. If the supply is capped this creates limited flexibility in responding to market demands, and has the potential for hoarding, as holders may not want to burn due to scarcity. However, we recognise that inflation without rigorous oversight can still drift toward mission dilution. Here we could create a "Commons Treasury Council" of elected stewards who approve or reject inflation proposals based on real-time ecosystem metrics. We can also consider **hybrid logic**: semi-capped with elastic

perimeters, unlocked only via DAO-triggered consensus under clearly defined emergencies or milestones.
These adaptive features could be triggered under strict governance protocols by, for instance: Emergency response needs (e.g. disaster fund, sudden censorship crisis), network expansion milestones (e.g. onboarding 10,000 verified contributors), or campaign triggers (e.g. global mobilisations, urgent aid funding).

Example Use Case for HRDAO:

Default State: The utility token (HRT-UTL) has a soft cap of 100 million tokens.
Trigger Event: A sudden internet blackout in Gaza prompts urgent response needs.
A DAO proposal is raised: Mint 1M extra tokens into a Solidarity Reserve Pool to fund encrypted mesh hardware, activist stipends, and zK onboarding kits.
Governance Condition:
At least ⅔ of active HRT-GOV holders vote yes,
Plus at least 10,000 HRT-UTL holders signal support through a burn-to-activate quorum model.
Release: Tokens are minted and distributed via multisig steward wallets with verified proof-of-need logs (linked to DePIN, on-chain requests, and campaign validators).

This adds elasticity with intention, not exploitation.
We can see a real world example of this adaptability in MakerDAO's Emergency Mint Switch, where in black swan events, MakerDAO has the ability to mint additional DAI to maintain peg stability. This is only triggered through extreme governance consensus and has strict caps and cooldowns. The elasticity is constrained by community panic controls, not just market signals.

Challenges with an inflationary model can also include a potential dilution of value for existing token holders if too many tokens are minted. The key challenge lies in designing an inflationary model that fuels growth and participation without undermining long-term trust or diluting the value held by early contributors.
To balance growth with value preservation, the HRT inflationary model should be **adaptive and purpose-driven**—tying new token issuance directly to measurable contributions, onboarding milestones, or ecosystem expansion metrics. Much of this can be achieved through crowd-sourced DAO participant consensus. Inflation should not be arbitrary; rather, it should reflect the DAO's real-world and on-chain evolution. A tiered emission schedule can also be introduced, where inflation decreases over time or adjusts dynamically based on participation rates and treasury health. Minting could be governed by collective intelligence—requiring quorum-based approval or algorithmic constraints triggered by key indicators like active users, proposal throughput, or tooling utilisation. This ensures that every newly issued token carries intent, legitimacy, and reciprocal value, aligning growth with stewardship and preventing runaway dilution.

Circulating Supply and velocity – How many tokens are actively in use?
Increased demand for a token can lead to higher transaction volumes, thus increasing velocity. Token Velocity = Total Transaction Volume / Average Token Supply. This calculation helps in assessing how effectively a token is being utilised within its ecosystem. Implementing staking mechanisms encourages users to lock up their tokens for a specified period, which reduces the number of tokens in circulation, thereby decreasing velocity. To effectively manage token velocity, the HRDAO must strike a balance between **circulation and commitment**. High velocity may indicate active usage, but unchecked, it can dilute the perceived value and long-term holding incentives. By

introducing **staking, time-locked rewards, and tiered access to services**, we encourage members to hold tokens with intention, reducing free-floating supply without stifling engagement. Velocity, in this sense, becomes a signal rather than a symptom—measuring the pulse of participation. Strategic utility design, such as requiring token access for high-value tools or enhanced governance participation, ensures that circulation is meaningful, not merely transactional. In this way, token velocity becomes an organic pulse, continually aligning with the DAO's evolving purpose.
We could also implement a sliding scale of "proof-of-care" scores tied to staked tokens, where not only quantity but *quality of engagement* boosts standing. Likewise vesting schedules should be public and traceable via a visual dashboard, to build transparency and public legitimacy.

Vesting Periods – What lock-ups apply for team or early backers? Token vesting and lock-up periods are essential mechanisms used to manage the distribution of tokens over time, ensuring that team members, advisors, and early investors do not burn their tokens all at once, which could negatively impact the token's market price. **Lock-up periods** for Amnesty team members and early partners are vital to protect the token's value and ensure long-term alignment. Tokens will be gradually released over time through a **vesting schedule**. This keeps everyone—founders, advisors, and early partners—committed to the DAO's future. While the exact structure will be defined later, the intention is to have an initial allocation with lock-ups in place, followed by minting and burning tied to actual DAO activity. This approach supports stability, builds trust, and rewards those who contribute over the long haul.

Interoperability – The blockchain ecosystem is growing increasingly interconnected, so it's important to take care of **interoperability at the design stage**. The HRDAO will have higher adoption chances, and more user engagement and popularity if it can interact with other blockchains without friction. The HRDAO is designed as an end-to-end, privacy-preserving, and governance-centric platform built on the Cardano blockchain, enhanced by Midnight's zero-knowledge layer, Apex Fusion's bridging and AI-driven agent support. We are encouraged by the inter-operable ethos of Cardano and other developing blockchain eco-systems.

Token Demand – Creating demand for tokens is crucial for maintaining their value and ensuring the sustainability of the project. Several strategies will be employed to generate demand within the HRDAO, including:
- Building a **Strong Community**: Engaging with users and fostering a vibrant community to create a loyal user base that actively participates in the ecosystem. Much of the existing thematic advocacy and campaigns data can be thoughtfully rerouted through the HRDAO.
- **Partnerships** and Collaborations: Collaborating with other stakeholders or human rights projects to expand the token's use cases and increase its visibility, driving demand both on and off-chain, digitally and IRL.
- **Awareness Campaigns**: Effective marketing strategies will educate potential users about the token's benefits and use cases, attracting new participants, and on-boarding curious and tech-savvy members.
- **Utility Expansion**: Continuously expanding the utility of the token by adding new features or use cases can keep the community engaged and attract new users, lifting the perceived value of the token. Of note here is the Tencent China model where users generate advertising and attention traffic by allocating reading time, steps or simple day to day activities. Inspired by Tencent's breakthrough in transforming everyday human activity — like reading or walking — into measurable, monetisable value, the HRT model can reimagine human rights advocacy as an ecosystem where **participation itself generates worth.** Instead of relying solely on donations or institutional action, we

can harness *human energy* — time spent engaging with rights content, participating in campaigns, building local knowledge, or merely the attention taken when walking or reading an aligned book or report — as a foundational currency. This turns rights support into a living, breathing economic system where value flows from dignity, attention, and care — a radical and empowering leap for both token utility and the future of global advocacy.

- **Feedback Loop**: As the token's utility value increases, it will attract more users, creating a positive feedback loop that further enhances its value. This phenomenon is often observed in blockchain token economies, where the growth of the user base and associated earn and burn metrics leads to increased token value (both in terms of value and potentially money).
- **Staking Rewards –** In proof-of-stake systems, users can earn tokens by staking their existing tokens, which helps secure the network while providing a financial incentive for users to hold onto their tokens. As more users stake their tokens, the network becomes more secure and resilient, attracting further investment and participation. Participants earn rewards in the form of additional tokens for their staking efforts, incentivising them to contribute to the network's health. This is a common practice in tokenised economy, and we will design a staking rewards system for the HRDAO.

Long-term Economic Sustainability and Growth: In the context of the HRDAO, long-term economic sustainability is not just about survival—it's about **stewardship of value** across generations. Our tokenomic model must be regenerative, rooted in the lived realities of global advocacy, and responsive to both on-chain metrics and off-chain movements. As the DAO evolves, mechanisms such as dynamic minting, mission-aligned staking, and adaptive treasury strategies will enable us to respond to shifting market conditions while preserving the token's integrity and purpose. By embedding resilience into the token's design, we ensure that value flows in alignment with justice—not speculation—and that the DAO remains a living, breathing engine for human rights, long after the launch hype has passed.

A well-balanced token supply model is essential for the Human Rights DAO to remain mission-focused and economically sound. A dynamic issuance schedule, shaped by real participation and community needs, could help the DAO evolve without locking it into rigid constraints. Vesting and gradual release timelines will ensure long-term alignment between early contributors, advocates, and the broader ecosystem. Ultimately, demand must be rooted in purpose. The token's value should grow from the real utility it offers—supporting human rights tools, rewarding action, and building trust—rather than from hype or short-term speculation.

"Demand comes from the token fulfilling a purpose, and being trusted to deliver on this purpose in a sustainable way." [25]

<u>Aligning Interests - Incentives and Participation:</u>

Incentive design is often where protocols rise or fall. Too often, tokenomics is reduced to simplistic reward schemes. HRDAO understands that incentives are psychological, social, and economic. Incentive structures must do more than encourage mere activity. They must reward meaningful participation and nurture a regenerative cycle of creation, contribution, and recognition. Holding tokens should bring both practical and emotional rewards—staking could enhance one's role in governance or unlock new collaborative opportunities. Participation itself becomes a form of investment in the collective.

"*Utility and interest alignment are key.*" [26]

Incentives should:

- Encourage long-term holding through staking and governance rights
- Reward meaningful contributions (not just activity)
- Foster decentralised participation gradually through distribution
- Distribute tokens in ways that are meritocratic and mission-aligned

Incentives should encourage not just participation, but meaningful contributions.

Some examples might include:

- Staking tokens to access advanced research or training.
- Incentivising consensus on the state of the network.
- Tokenised contributions for grassroots campaigns or artivism.
- Transparent funding for urgent human rights actions.
- Rewarding contributors for verified translations, documentation, or grassroots reporting.
- Reputation scores tied to contribution quality and engagement.
- Incentivising participants to act in a manner consistent with upholding human rights.
- Rewarding actions that advance advocacy for a human rights cause (let's say proof-of-advocacy).

The token economy becomes a mirror of the culture it supports—regenerative, empowering, and transparent, and HRT should act not only as a reward, but as a **catalyst for action, recognition, and governance**. Minting occurs when someone records or verifies a meaningful action—like documenting a human rights abuse, hosting a local advocacy event, capacity building on the skills platform, submitting evidence to a campaign, or offering peer-to-peer support. These tokens aren't passive—they represent *proof of care*, *proof of courage*, and *proof of contribution*.

But tokens also need gravity. A sink must exist to give weight to their flow. So, holders can burn tokens to **vote on policy priorities**, **fund a local journalists' work, support an innovative artists project or urban solution, trigger urgent DAO responses**, or **initiate restorative justice circles** within the network. This balances abundance with accountability.

The token flow becomes cyclical: *create → engage → steward → transform*. For example:

- **Whistleblowers** mint tokens when verified by a trusted node.
- **Researchers or legal advocates** receive tokens for contributing to a decentralised case archive.
- **Artists and storytellers** are rewarded for shaping narratives that inspire action.
- **Journalists** receive tokens for geo-spacial verified human rights reports from the field.
- **Burning tokens** to support the above efforts, or to unlock temporary roles in sub-DAOs, allow deeper access to sensitive platforms, or activate conflict resolution tools.

As Sherman (2020) says:
"purpose-driven tokens provide an alternative to conventional economic systems, which predominantly incentivize individual value creation ... however this new and collective value creation phenomena ... will likely need much more research and development, and a long phase of trial and error, before we can better understand the potential of incentivizing contributions to a public good." [27]

The Tencent model in China [28] (referred to above), pioneered a form of "**attention economics**" where users contribute value simply by engaging with everyday activities — reading, walking, sharing — and these interactions are directly monetised or tokenised. This model transforms passive participation into an active economic engine, redistributing micro-value flows back to individuals. It's a reimagining of social energy: time, steps, and attention become ledgered forms of labour and influence.

For the Human Rights Token (HRT), this opens up a **revolutionary design space**: what if *acts of citizenship* — from signing petitions, attending community forums, to simply reading educational materials on human rights — were valued, and tokenised? This would allow people around the world, especially those in marginalised contexts, to *contribute to and benefit from* the human rights movement through their everyday actions. It decentralises participation and makes the movement more resilient and inclusive, whilst bringing attention organically to worthwhile human rights causes, advocacy opportunities and action. We could also actively encourage active participation as a recurring gamified challenge. This opens up the space for **real-world "rituals"** (walking, reading, podcasting) to generate value—this redefines what counts as *labour* and *contribution*, honouring embodied participation. Oracle design and validation logic for verifying off-chain engagement (e.g. reading, walking) needs further investigation. **Mutuality and agency** are here key.

Utility expansion, then, becomes more than feature creep — it is a strategic reinvention of value itself. By aligning token utility with embodied, day-to-day engagement, we tap into a new form of advocacy economics: a web of living labour generating rights-based value, rooted not in extraction but in recognition. Just as these models reward steps, HRT can honour the steps taken — metaphorically and literally — towards human rights justice.

Suggestions for enhancement include:

- Foster a ritualised burn protocol, e.g. token sacrifices to memorialise fallen activists or fund emergency actions, embedding symbolic depth.

  > Scenario: "As I was in Berlin doing my artivist trail, with a fellow Jedi web3 cultural engineer and web3 multichain OG builder, I went to the Jewish Museum wearing my keffiyeh to take photos. We contemplated on the history of the gas chamber and holocaust, and thought about tokens, gas, burning tokens, and I thought about the act of burning tokens could be done in specific places with significance and meaning to unlock certain tokens and artivist digital artefacts."

- Make the oracles community-curated by implementing a decentralised network of "trusted nodes" - a rotating cohort of DAO-recognised contributors such as human rights activists, frontline witnesses, artists, researchers, and cultural workers—who collectively verify and contextualize proofs-of-action submitted by participants.
- These trusted nodes operate not as gatekeepers, but as peer witnesses in a pluralistic validation system rooted in care, presence, and cultural literacy. Each action—whether it's a walk for Gaza, a report from Ukraine, a solidarity poem performed in Paris, or the mapping of a destroyed site in Sudan—is not just reviewed by algorithms or GPS. It is verified by humans who understand the stakes.

Contextual Example: Palestine/Gaza
Scenario:
A Palestinian activist in Rafah uses her HRDAO dApp to document a destroyed school using OpenStreetMap tagging. She adds a brief description and uploads a short voice recording.
Step 1:
A DePIN-connected sensor node (part of a trusted node registry), confirms her location and environmental metadata (e.g., time, air quality).
Step 2:
A trusted node in Beirut, formerly a civil society educator, reviews the post. She confirms the coordinates and adds a tag: *"Destruction of educational infrastructure // Rafah // 2025"*
Step 3:
An artist-node in Ramallah adds a symbolic glyph and shares the report to a DAO-curated Solidarity Map.
Step 4:
The DAO dashboard now lists this proof as "Verified x3," earns the witness contributor 100 HRT-UTL, and escalates the case to the governance track for campaign activation.

In essence, the Human Rights Token becomes a **cultural and functional currency**—it reflects the moral human rights economy of the Amnesty HRDAO. One that grows not by extraction, but through the constant interplay of witness, response, action and collaboration.

"The utility of a token is a key determinant of its value. Tokens with high utility have more demand as they can be used for various purposes within their ecosystem." [29]

Tokenomics Conclusion: A Covenant, Not a Contract

We regard tokenisation as not only representing assets within the Web3 ecosystem but as also embodying a commitment to social justice. That is, tokens as tools to catalyse moral and societal change. In our case with Human Rights.

It allows us to ask not just *what is the value of this token?*—but *what values does this token represent?*

In this, we move beyond economics into ethics. And in doing so, we build a DAO not only worthy of Amnesty International's legacy, but of the global human rights embodied future we are daring to imagine.

The HRDAO's token economy is a kind of constitutional layer. A coded covenant between members demonstrating that participation matters and that solidarity can be transacted as a value system. In this way we can embed mutuality and agency at the core of the design for the HRDAO.

"If a lot of people want to contribute [to a DAO] in order to change something for the better, that motivates others to do the same, creating a positive feedback loop that DAOs are perfectly suited to capitalize on." [30]

It's important to note that the Human Rights Token (HRT) is *not* a speculative asset. It is a **non-extractive governance tool**—a mechanism for collective decision-making, activism coordination, and regenerative funding within the Amnesty Human Rights DAO. By avoiding speculation, HRT also reduces exposure to the growing regulatory scrutiny facing tokens framed as financial instruments—protecting both the DAO and its members from legal risk while preserving its activist mandate.

HRT derives its value not from market volatility but from its **utility in organising people power**. It is a token of shared responsibility, not financial gain. Through it, grassroots communities, activists, and aligned networks can vote, fund, and govern campaigns with transparency, accountability, and purpose. Rather than replicating extractive crypto models, HRT supports a **pluralistic and decentralised rights movement**, and encodes **solidarity into protocol**, making the DAO not just a tech stack—but a living social covenant.

The **Human Rights Token (HRT)** is more than a digital artefact—it is a vessel of *collective intention*. Born not from market speculation, but from movement logic, it exists to enable global coordination, fund frontline action, and uphold dignity through participatory governance. HRT is *mission-pegged*, not market-pegged; its value is rooted in solidarity, trust, and purposeful engagement—not price charts or trading volume.

HRT proposes a **regenerative** dual-token architecture where **Utility Tokens** reward contribution and unlock access, and **Governance Tokens**—soulbound by design—reflect enduring trust, reputation, and commitment to collective stewardship.

Use Case One: Signalling Support Through Earned or Donated Value for Human Rights Projects

In the HRDAO ecosystem that honours participation as a form of agency, individuals can express support for human rights projects—whether they manifest as for instance activist campaigns or artistic expressions—through a unique value signalling mechanism powered by the HRT. Participants earn HRT through everyday interactions with the DAO: voting on proposals, contributing to discussions, curating knowledge, or completing meaningful tasks. But beyond traditional DAO functions, the token model is extended into the broader attention economy. Here, time itself becomes a form of capital—users earn tokens by listening to human rights podcasts, raising their voice in human rights contexts, reading investigative journalism, or even walking while meditating on the project's purpose, potentially self-nominated and verified, or tracked via device-integrated oracles.

This system allows people who may not have financial capital to meaningfully contribute by donating "proofs of attention"—a metric that could include time spent on a campaign's materials, cognitive energy expended during a collective brainstorm, or embodied movement such as steps taken in solidarity. These contributions are converted into micro-value units that can be staked or donated to specific projects or campaigns. The symbolic and economic gestures converge: the token becomes a digital vote of confidence, a cultural 'tip' of endorsement, and a quantifiable source of attention and moral investment.

Each donation not only strengthens a project's perceived importance within the DAO's decision-making algorithm but may also unlock further matched funding, reputation boosts for creators, or collaborative invitations for donors. The system supports cross-border value exchange without centralised mediation, honouring the pluriversal logic of global human rights in our evolving futures.

Token Flow:

- **Input:** User engages in DAO participation (voting, curating, reading articles, walking in solidarity, listening to a podcast).
- **Value Accrual:** System tracks engagement through proof-of-attention or physical proof-of-action (verified oracle needed).
- **Token Minting:** User receives HRT based on weight of interaction.

- **Signal Use:** User donates HRT to a chosen human rights project (e.g., art exhibit, protest documentation, educational campaign), or burns it to signify alignment with a campaign or activity.
- **Feedback Loop:** Project with higher token backing gains visibility, potential DAO matching funds, and access to more collective resources.
- **Outcome:** Creates an ecosystem where ethical attention and civic participation are transacted as cultural value.

<u>Use Case One Scenario: Signalling Support Through Donated or Earned Value</u>

Characters: Alicia (DAO contributor in Nairobi), Bob (educator in Birmingham), Raz (artist-activist in Berlin).

Alicia is an active participant in the Human Rights DAO. She regularly contributes to the governance forums—voting on proposals, curating trusted sources for the knowledge hub, and tagging verified campaign materials. Over the past week, she's earned HRT tokens for these activities. Today, she's browsing the DAO's "Campaign Spotlight" and discovers Raz's project: an augmented reality exhibit in Berlin documenting the daily lives of undocumented delivery workers.

While inside the DAO, Alicia reads the campaign summary and browses its attached proposal PDF. That reading time is logged as a weighted interaction. At the end, she's prompted to donate HRT to support the project's growth—she contributes 40 tokens she earned from her participation. She then links her external activity: she spent an hour walking while listening to Raz's podcast on migration and digital borders. Her step-tracked engagement is submitted to the DAO via an off-chain oracle, and after verification, she's rewarded with a small HRT top-up.

Bob, meanwhile, is reviewing DAO educational resources to prepare a class module on civic engagement. His time spent inside the DAO reading the human rights curriculum earns him HRT. When he sees that Raz's campaign is trending in the DAO interface, boosted by Alicia's donation and public attention metrics, he donates 25 of his tokens too. This push crosses the "collective signal threshold," unlocking matched DAO treasury support and giving Raz's project eligibility for a DAO-hosted workshop with migrant organisers. The economy of support—reading, walking, listening, donating—creates a feedback loop of cultural legitimacy and funding.

Token Flow Recap:

- **DAO Activity:** Alicia earns HRT by voting, curating, and reading proposals.
- **Public Activity:** Alicia walks while listening to a project-aligned podcast. Oracle logs steps + attention.
- **Token Use:** Alicia donates tokens to Raz's campaign from both on-chain and off-chain activities.
- **DAO Mechanism:** Bob donates tokens after discovering the campaign through the DAO dashboard.
- **Outcome:** Project crosses donation threshold, triggering DAO matching funds and increased visibility.

Use Case Two:
Field Reporting and Secure Verification by Activists, Migrants, or Journalists

Imagine a frontline activist, displaced migrant, or independent journalist operating in a high-risk environment—conflict zone, refugee corridor, or authoritarian regime. Traditional media channels may not trust or be able to verify their report. Through the HRT ecosystem, these actors can use token-based access to submit securely encrypted, geo-stamped reports—be they text, video, or voice—on a distributed ledger anchored in verified identity frameworks like SILT. This ensures both authentication and protection of the source, all while preserving anonymity where necessary.

Access to this submission portal is gated via earned or donated HRT tokens, which both ensures the seriousness of interaction and discourages spam or bot manipulation. Once submitted, the DAO members and aligned civil society groups can amplify the report by staking their own tokens as a trust signal, boosting visibility, and increasing credibility. In essence, token staking functions as a decentralised editorial layer, affirming what's worthy of attention and solidarity. This also invites journalists and civil society groups to curate verified field updates into knowledge products (policy briefs, art, exposés) with collective legitimacy.

This model allows for real-time solidarity and reputation-building: a field report may spark an immediate campaign, policy response, or aid intervention. It also provides journalists and activists with a decentralised credentialing system, allowing their fieldwork to accumulate a kind of social capital that can be later redeemed as access to funding, platforms, or digital safe havens. It collapses the space between testimony and action, report and mobilisation—empowering the margins to speak with authority.

Token Flow:

- **Input:** Field actor (activist, journalist, migrant) obtains HRT via grant, donation, or DAO participation.
- **Access:** Uses HRT to unlock secure reporting portal with GPS/metadata verification and optional anonymisation tools.
- **Submission:** Uploads verified report (audio, video, text).
- **Amplification:** Other users stake tokens to signal trust and boost visibility.
- **Validation:** Token-backed reports are curated and shared with aligned networks (media, NGOs, DAOs).
- **Outcome:** Real-world solidarity and action driven by decentralised verification and tokenised trust signals.

Use Case Two Scenario: Secure Reporting and Token-Gated Amplification

Characters: Marwan (field journalist in Tripoli), Eva (digital rights researcher in Brussels), Raz (again, as signal-booster)

Marwan, working as an independent journalist in Tripoli, captures video and audio of state-sanctioned evictions of migrants from temporary shelters. Instead of uploading to a public site, he logs into the Human Rights DAO and uses his verified journalist governance token enabled by zK pseudo-anonymity to access the secure field-reporting portal. His wallet verifies access, and the portal encrypts and geostamps his footage while allowing for metadata obfuscation to protect his identity. He attaches a short description and tags the submission under the “Displacement” report stream.

Eva, monitoring the DAO’s verified field feed, spots Marwan’s submission. She’s already earned HRT by editing DAO documentation and reviewing submitted reports for clarity. She reviews Marwan’s report inside the DAO platform, confirms its integrity through metadata, and stakes 100 of her tokens to boost its visibility within the DAO’s “Urgent Reports” dashboard. Her staking is both a credibility signal and a reputational endorsement—DAO algorithms now flag the post for potential public release and embed it in a weekly intelligence digest.

Raz, who regularly scans DAO-verified reports for creative material, views Marwan’s footage after it trends in the DAO’s feed. He uses 10 of his own HRT to generate a public campaign visual: an info-graphic version of the report that can be shared externally while preserving its on-chain verification mark. He embeds it into a projection piece for a Berlin street action and directs donations from QR code scans back to the journalist protection pool within the DAO. This creates a virtuous loop: verified field reports → curated DAO signal → public visibility → new HRT donations → protection for frontline witnesses.

Token Flow Recap:

- **DAO Access:** Marwan uses earned or donated HRT to submit a secure report (encrypted, GPS-tagged).
- **DAO Engagement:** Eva reviews and stakes tokens to amplify the report’s visibility and trust.
- **DAO to Public:** Raz visualises the report through DAO tools, embeds it in a public artwork.
- **Public Impact:** External users scan a QR code, send new donations into the DAO pool for field reporters.
- **Outcome:** Field report gains reach, the journalist remains protected, and a cycle of trust and funding is reinforced.

These use cases present a layered, fluid, and deeply ethical interaction between attention, agency, and activation—embedding the Human Rights Token within the fabric of cultural, political, and spiritual life.

These scenarios don't just demonstrate utility—they narrate a new form of digital-social alchemy. Everyday actions—reading, walking, voting, curating—are transformed into real-world solidarity through the Human Rights Token. They exemplify a shift away from the mentality of *“what can one individual do?”* towards an individual that participates, protects, and uplifts through collective technology.

To realise its full potential, this model will need radical clarity in onboarding and UX, strengthened systems of verification, ritual, and accountability, and coherent narratives across education, governance, and participation.

8 Conclusion: Towards Regenerative Governance for Human Rights

This whitepaper lays the foundational architecture for a governance and token system capable of transforming Amnesty International's mission into a decentralised, participatory, and durable DAO structure — one that evolves with the communities it serves. What emerges is not merely a governance framework, but a living instrument—rooted in real-world activism, informed by interdisciplinary research, and designed for responsible scalability.

Through a comparative lens, we've examined DAO governance models that honour transparency, pluralism, and community control. We've grappled with the hard edges of token design, surfacing key challenges such as meaningful participation, potential misuse, and speculative dynamics. And we've rooted our vision in lived realities — with practical use cases aligned with Amnesty's campaigns and deliverables, ensuring our framework is not merely theoretical, but operational and impactful.

In response to the interwoven challenges of distrust, marginalisation, and systemic rigidity that persist across both civil society and emerging decentralised systems, this initiative does not claim to resolve complexity through finality. Rather, it commits to a governance architecture that is iterative, inclusive, and responsive—one that distributes agency without disintegration, anchors accountability without reverting to hierarchy, and makes space for diverse worldviews without collapsing them into uniformity. Anchored in the Cardano ecosystem, HRDAO leverages protocol-level tools—SIP-1694, Voltaire, Atala PRISM—not as technological ends, but as relational enablers: instruments in service of collective justice.

At the core of this paper lies a set of deliberate innovations designed to bridge the technical and the ethical. The dual-token architecture—distinguishing between utility and soulbound governance tokens—protects democratic integrity while rewarding genuine contribution. Quadratic and reputation-weighted voting mechanisms counteract plutocracy and enable equitable participation across geographies and lived experiences. By integrating zero-knowledge proofs, Atala PRISM credentials, and a Council of Stewards, HRDAO supports secure engagement for those at risk and embeds human rights assessments into the governance substrate itself. This is not speculative architecture—it is purposeful, actionable infrastructure—designed to uplift grassroots voices, decentralise decision-making, and encode solidarity into the chain's very logic.

This is not the end of the conversation, it is merely a gateway. What we have presented here is a provocation for new forms of organising global solidarity, particularly in terms of social justice, human rights and impact. A first milestone in a journey to encode human rights into systems that are borderless, trustless, and co-governed by the people they aim to protect.

To prospective allies, funders, and supporters: this is an invitation. We are building not just a tool, but a new institution for the 21st century — a Human Rights DAO that anchors our shared values in cryptographic trust, inclusive governance, and global action.

The path ahead will require care, experimentation, and principled collaboration. Yet with the clarity of vision laid out in this whitepaper, and the active support of the Project Catalyst community, HRDAO stands poised not merely as a prototype—but as precedent. A new chapter in Cardano's governance evolution, where human rights are not only defended, but enacted, encoded, and continually renewed in the decentralised age.

9 Notes and References:

Notes:

[1] Unlocking New Horizons – Exploring Blockchain Technology in Human Rights Education and Supporter Coordination for the Global Amnesty Movement. Retrieved from: https://cdn.sanity.io/files/ysiap3nf/production/0c128a5330f180f206e63791f21b69f916ee19d8.pdf

[2] Ohlhaver, Puja, E. Glen Weyl, and Vitalik Buterin. "Decentralized Society: Finding Web3's Soul." *SSRN*, May 10, 2022

[3] refer https://milestones.projectcatalyst.io/projects/1300018

[4] Sherman (2020) pg 120

[5] Owoko, K. and Puncar (2025) pg 6

[6] De Filippi, et al (2024) preface pg x

[7] Refer general frameworks like https://www.gida-global.org/care or culturally specific frameworks like https://www.temanararaunga.maori.nz/

[8] Werbach (208) p. 134

[9] ibid p. 135

[10] DAOs - The New Coordination Frontier. Comprehensive report curated by Bankless and Gitcoin DAOs 2021 edition

[11] De Filippi, et al (2024) pg 122

[12] Ostrom (1990)

Elinor Ostrom's 8 Principles for Governing the Commons:

- Clearly Defined Boundaries
  The resource and its user group must have clear boundaries—who has rights to use the resource must be known and limited.
- Congruence Between Appropriation and Provision Rules and Local Conditions
  The rules for using resources should fit local needs and conditions. Benefits and responsibilities should be proportional to the amount of use.
- Collective-Choice Arrangements
  Most individuals affected by the rules can participate in modifying them—this ensures rules are adapted to actual lived realities.
- Monitoring
  Monitors, who actively audit resource conditions and user behaviour, are accountable to the users or are users themselves.
- Graduated Sanctions
  Rule violations are sanctioned in a graduated way, depending on the seriousness and context of the offence—not overly harsh, but not ignored.

- Conflict-Resolution Mechanisms
  There must be accessible, low-cost means for resolving conflicts between users or between users and officials.
- Minimal Recognition of Rights to Organise
  The rights of users to self-organise and create their own institutions must be respected by external governmental authorities.
- Nested Enterprises *(for CPRs that are part of larger systems)*
  In larger systems, governance activities are organised in multiple layers of nested enterprises—small local units within larger regional or national frameworks.

These principles were radical in challenging the binary of "tragedy of the commons" vs. state control, showing that communities can self-organise effectively—given the right conditions. Ostrom's work reframes governance as a polycentric, co-designed process rather than a top-down imposition. It's foundational for regenerative design, DAO logic, and pluriversal systems thinking.

[13] rf Zoltu, M., & Davidson, S. (2023). *The Political Economy of DAOs: Coordination Challenges, Economic Incentives, and Governance Design*

[14] as discussed in Ding, Q., Xu, W., Wang, Z., & Lee Kuo Chuen, D. (2023). *Voting Schemes in DAO Governance. Singapore University of Social Sciences.*
Where Eight primary voting schemes are analysed:

- Token-based quorum voting: Requires a certain percentage of voter participation for a proposal to be valid.
- Quadratic voting: Balances voting power between large and small stakeholders.
- Weighted & Reputation-based voting: Assigns more voting power to experienced or invested members.
- Knowledge-extractable voting: Rewards informed decision-making.
- Multisig voting: Requires multiple signatories to approve proposals.
- Holographic consensus: Uses predictive markets to streamline governance.
- Conviction voting: Allows continuous staking on proposals, favouring long-term commitments.
- Rage Quitting voting: Enables dissenters to exit with their share of DAO assets.

[15] cited in Werbach (2018), pp. 135–136.

[16] ibid note [12] above

[17] Council of Europe Overview Report (2022) pg 16

[18] ibid

[19] rf Schneider 2022

[20] ibid

[21] ibid pg 123

[22] Sherman (2020) pg 324

[23] Lacity (2022) pg 100

[24] Piech (2024) pg 21

[25] ibid

[26] ibid pg 20

[27] Sherman (2020) pg 268

[28] https://www.tencent.com/en-us/about.html

[29] Frank (2023) pg 25

[30] Owoko, K. and Puncar (2025) pg 63

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